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ECM617 - RADIO FREQUENCY SYSTEM DESIGN

TECHNICAL REPORT

MINI PROJECT: FILTER DESIGN

GROUP: CEEE222S7A

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1.0 INTRODUCTION

This project concentrates on the design of a microstrip interdigital Chebyshev filter intended for 5G applications. The Chebyshev filter is commonly chosen in RF systems because it provides a steeper roll-off compared to maximally flat responses, allowing for improved frequency selectivity near the passband edges. Although a controlled ripple is introduced in the passband, this trade-off enables stronger suppression of unwanted signals outside the operating band, which is crucial for dense 5G and LTE environments. Such characteristics make the Chebyshev response well suited for modern wireless systems that demand efficient spectrum utilization and high interference rejection. Furthermore, the Chebyshev filter can be readily implemented using microstrip technology, with interdigital structures offering compact size and ease of fabrication. By exploiting the sharp transition and predictable behavior of the Chebyshev response, the proposed filter design achieves low insertion loss within the passband while maintaining adequate stopband attenuation to satisfy the performance requirements of contemporary 5G communication systems.

2.0 METHODOLOGY

2.1 Specification Definition

The first step of the design process involves defining the electrical and physical specifications of the microstrip interdigital filter based on the requirements of 5G mid-band applications. A center frequency of 3.5 GHz is selected, which is widely used in 5G communication systems. A Chebyshev bandpass response is chosen due to its sharper roll-off characteristics and improved selectivity compared to maximally flat filters.

The filter is designed to achieve 20 dB attenuation at frequencies ± 0.4 GHz from the center frequency, ensuring effective suppression of adjacent channel interference. The substrate material used is FR4, with a relative dielectric constant (ϵ_r) of 4.5, substrate thickness of 1.6 mm, and loss tangent of 0.017, representing a low-cost and widely available PCB material.

2.2 Selection of Filter Type and Order

A Chebyshev bandpass filter is selected for this design due to its superior frequency selectivity compared to maximally flat responses. This characteristic is achieved by allowing a controlled ripple in the passband, which results in a sharper roll-off and improved attenuation near the band edges. Such performance is highly desirable for 5G applications at 3.5 GHz, where adjacent channel interference must be effectively suppressed.

In this work, a passband ripple of 0.2 dB is chosen as a compromise between insertion loss and stopband attenuation. This ripple level provides sufficiently steep skirts to meet the requirement of 20 dB attenuation at frequencies $f_0 \pm 0.4$ GHz, while maintaining acceptable passband performance.

The required filter order is determined using standard Chebyshev filter design procedures based on the specified ripple level and attenuation requirement. Using the normalized low-pass Chebyshev prototype table for 0.2 dB ripple, the corresponding prototype element values ($g_1, g_2, \dots, g_n$) are selected. These values form the basis for transforming the low-pass prototype into a bandpass response centered at 3.5 GHz.

The selected filter order ensures that the design meets the desired selectivity and attenuation specifications while keeping the interdigital microstrip structure compact and suitable for implementation on an FR4

substrate.

(c) 0.2 dB ripple									
1	0.4342	1.0000							
2	1.0378	0.6745	1.5386						
3	1.2275	1.1525	1.2275	1.0000					
4	1.3028	1.2844	1.9761	0.8468	1.5386				
5	1.3394	1.3370	2.1660	1.3370	1.3394	1.0000			
6	1.3598	1.3632	2.2394	1.4555	2.0974	0.8838	1.5386		
7	1.3772	1.3781	2.2756	1.5001	2.2756	1.3781	1.3772	1.0000	
8	1.3804	1.3875	2.2963	1.5217	2.3413	1.4925	2.1349	0.8972	1.5386
9	1.3860	1.3938	2.3093	1.5340	2.3728	1.5340	2.3093	1.3938	1.3860

Figure 1: Normalized low pass Chebyshev filter

2.3 Low-Pass Prototype and Frequency Transformation

The filter design begins with a normalized low-pass Chebyshev prototype, where element values (g-parameters) are obtained based on the chosen ripple level and filter order. These prototype values are then transformed into a bandpass response using frequency scaling techniques.

The fractional bandwidth (FBW) is calculated using the lower and upper cutoff frequencies defined by the 20 dB attenuation points. This transformation allows the low-pass prototype to be mapped into a bandpass configuration centered at 3.5 GHz.

$$\begin{aligned}
 f_x &= f_0 + 0.4 \text{ GHz} \\
 &= 3.5 \text{ GHz} + 0.4 \text{ GHz} \\
 &= 3.9 \text{ GHz}
 \end{aligned}$$

$$\begin{aligned}
 w' &= \left(\frac{w}{w_c} \right) = \frac{(2\pi)(3.9 \text{ GHz})}{(2\pi)(3.5 \text{ GHz})} \\
 &= 1.114
 \end{aligned}$$

Formula Ripple and Number of Order:

$$\begin{aligned}
 \text{Ripple} &= 10 \log_{10} (\epsilon^2 + 1) \\
 0.2 &= 10 \log_{10} (\epsilon^2 + 1) \\
 0.02 &= \log_{10} (\epsilon^2 + 1) \\
 (\epsilon^2 + 1) &= 1.047 \\
 \epsilon^2 &= 0.047 \\
 \\
 L_q &= 10 \log_{10} [1 + \epsilon^2 \cosh^2 (n \cosh^{-1}(\omega'))] \\
 15 &= 10 \log_{10} [1 + 0.047 \cosh^2 (n \cosh^{-1}(1.114))] \\
 1.5 &= \log_{10} [1 + 0.047 \cosh^2 (n \cosh^{-1}(1.114))] \\
 31.623 &= 1 + 0.047 \cosh^2 (n \cosh^{-1}(1.114)) \\
 \\
 \frac{31.623 - 1}{0.047} &= \cosh^2 (n \cosh^{-1}(1.114)) \\
 \sqrt{651.553} &= \cosh (n \cosh^{-1}(1.114)) \\
 25.526 &= \cosh (n \cosh^{-1}(1.114)) \\
 n &= \frac{\cosh^{-1}(25.526)}{\cosh^{-1}(1.114)} \\
 n &= 8.313 \\
 n &\approx 9
 \end{aligned}$$

2.4 Interdigital Microstrip Structure Design

The bandpass filter is implemented using a microstrip interdigital configuration, which consists of parallel resonators coupled through edge coupling. Each resonator is designed to have an electrical length of approximately $\lambda_g/4$ at the center frequency, where λ_g is the guided wavelength on the FR4 substrate.

The resonator widths are calculated to achieve a characteristic impedance suitable for microstrip implementation, while the spacing between resonators is adjusted to control the coupling coefficient. Input and output tapping positions are optimized to achieve impedance matching and minimize return loss.

2.5 Substrate and Microstrip Parameter Calculation

Microstrip line dimensions are calculated using standard transmission line equations, taking into account the substrate dielectric constant and thickness. The effective dielectric constant (ϵ_{eff}) is determined to

estimate the guided wavelength and resonator length accurately.

Loss mechanisms such as dielectric loss and conductor loss are considered due to the relatively high loss tangent of FR4. These losses are expected to slightly increase insertion loss, especially at 3.5 GHz, and are accounted for during simulation.

2.6 Simulation and Optimization

The initial filter layout is simulated using an RF electromagnetic simulation tool such as Keysight ADS, AWR Microwave Office, or HFSS. S-parameters including S11 (return loss) and S21 (insertion loss) are analyzed to verify that the filter meets the design requirements.

Parametric optimization is performed by adjusting resonator lengths, widths, and coupling gaps to ensure the center frequency is accurately positioned at 3.5 GHz and that the 20 dB attenuation points occur at ± 0.4 GHz from the center frequency.

2.7 Performance Evaluation

The final design is evaluated based on key performance metrics, including insertion loss in the passband, return loss, bandwidth, and stopband attenuation. The Chebyshev response is verified by observing the passband ripple and steep roll-off characteristics. The obtained results are compared with the initial design specifications to confirm the suitability of the filter for 5G applications operating at 3.5 GHz.

3.0 RESULT AND DISSCUSION

3.1 Low Pass Filter

Calculating a low-pass prototype is essential in filter design because it simplifies the process by providing a normalized reference that can be transformed into bandpass, high-pass, or band-stop responses.

3.1.1 Calculation Low Pass Chebyshev

Formula used for L and C of low pass:

$$C = \frac{g}{Z_g W_c} \quad L = \frac{g Z_g}{W_c}$$

$$W_c = 2\pi f$$

$$= 2\pi (3.56)$$

$$= 21.9916$$

Calculation:

C1 = C9	G1 = G9 = 1.3860 $\frac{1.3860}{50 (21.991G)} = 1.261pF$
C3 = C7	G3 = G7 = 2.3093 $\frac{2.3093}{50 (21.991G)} = 2.1pF$
C5	G5 = 2.3728 $\frac{2.3728}{50 (21.991G)} = 2.15pF$
L2 = L8	G2 = G8 = 1.3938 $\frac{(1.3938)(50)}{21.991G} = 3.169nH$
L4 = L6	G4 = G6 = 1.5340 $\frac{(1.5340)(50)}{21.991G} = 3.488nH$

3.1.2 Result Low Pass Filter

A. Circuit Low Pass Filter

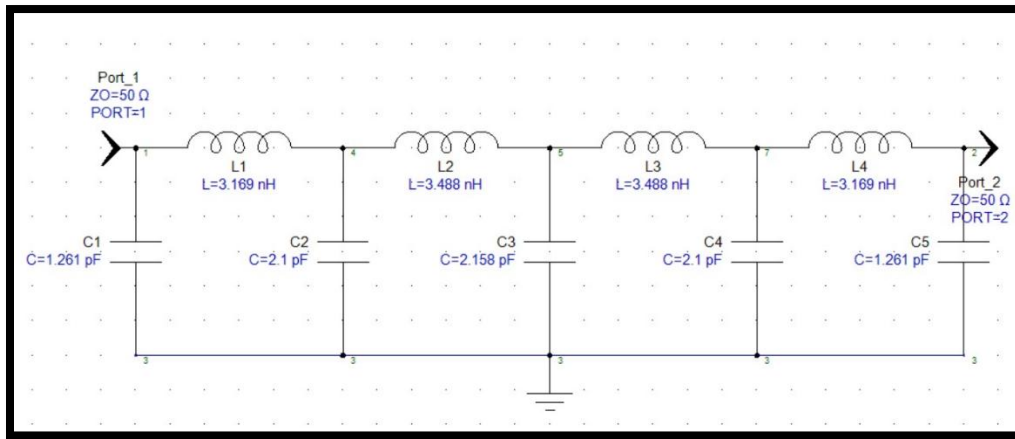


Figure 2

Figure two illustrates a lumped-element low-pass filter (LPF) that is designed as a 50 Ω input/output network of inductors and capacitors, which is a ladder network. The inductors (L1-L4) are placed in a series along the signal line, where the capacitors (C1-C4) are attached between the signal line and ground. This arrangement permits low frequency content to be carried and cuts off high frequency content. At low frequencies, the inductors are very small and the capacitors act like open circuits to a large extent, hence the signal is passed to the output at a minimal loss. The higher the frequency, the higher the inductive reactance and the lower the capacitive reactance, so the higher the frequency, the higher the tendency of the inductors to block the high frequency signals and the capacitors to shunt the high frequency to Earth. The additional LC sections enhance the filter order that leads to a better transition between the passband and stopband and higher attenuation of unwanted high-frequency signals. Since the values of the components fall in the nanohenry (nH) and picofarad (pF) range, this filter is evidently not aimed at low frequency electronics, but at RF or microwave use at the GHz range.

B. Graph

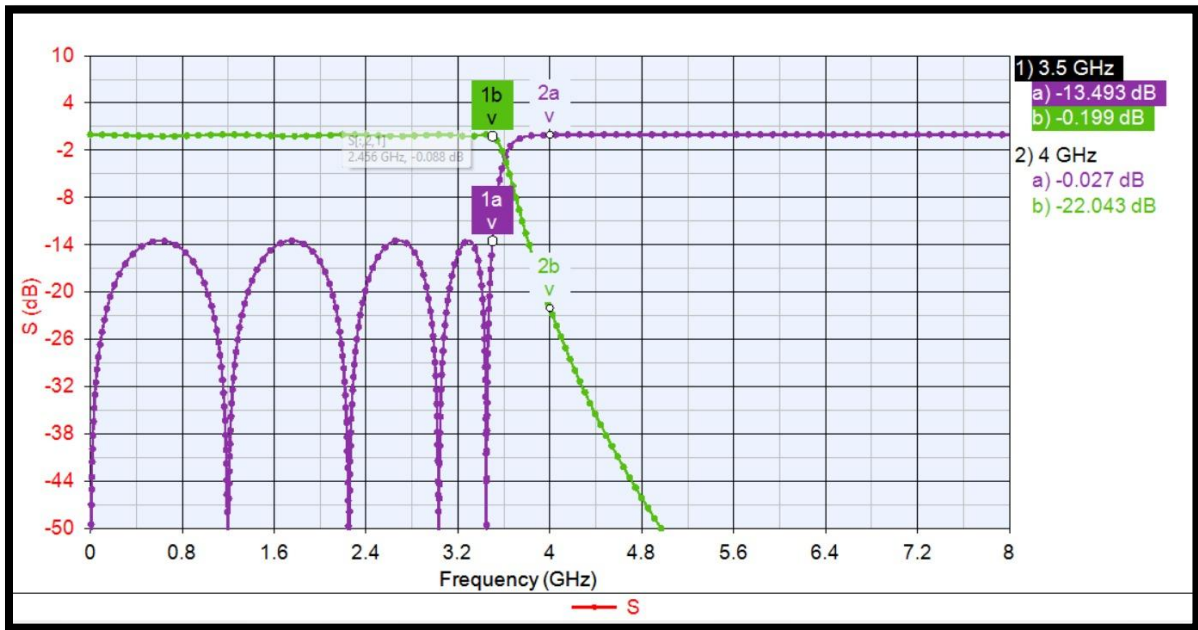


Figure 3: Frequency Response of a 9th- Order Chebyshev Low Pass Filter at 3.5GHz

Figure three which indicates the S-parameter performance of the low-pass filter. The frequency is indicated along the horizontal axis in GHz, and the magnitude is indicated along the vertical axis in decibels (dB). S21 curve (insertion loss) is the signal power which is passed on between the input and output. At the low frequency (passband), S21 is near 0 dB, i. e. there are minimal attenuation in passing the signal. Since it tends towards the cut off region (often a few GHz) the response starts to roll off steeply showing the transition band of the filter. Anywhere above the cut off frequency, S21 magnitude decreases rapidly to significant negative values (-20 dB, -40 dB, etc.), which shows that the attenuation in the stopband is high. The resonance effect of the LC sections causes the deep notches which are seen in the stopband and are characteristic of filters of higher order. On balance, the graph shows that the circuit in the first picture is acting as a proper low-pass filter, admitting the low-frequency signal components on its way but effectively suppressing the high-frequency components.

Parameter	Value(dB)
Return Loss, S_{11}	-13.493
Insertion Loss, S_{21}	-0.199

3.2 Transformation from Lowpass to Bandpass

Transforming from a lowpass to a bandpass prototype is necessary because the lowpass serves as a normalized starting point with well-defined mathematical properties to scale and shift it to the desired bandpass frequency range. The Butterworth response ensures a maximally flat passband, and by applying frequency transformations, this characteristic is preserved while adapting the filter to reject out-of-band signals.

3.2.1 Calculation

Formula used for series inductor:

$L_{B1} = Z_o \left(\frac{g_1}{\omega_1 \omega_2} \right)$	$C_{B1} = \frac{1}{Z_o} \left(\frac{\omega_1 \omega_2}{\omega_{r2} g_L} \right)$
---	---

Formula used for shunt low-pass capacitors:

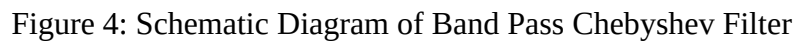
$L_{B1} = Z_o \left(\frac{\omega_2 - \omega_1}{\omega_{r2} g_c} \right)$	$C_{B1} = \frac{1}{Z_o} \left(\frac{g_c}{\omega_2 - \omega_1} \right)$
---	---

Calculation:

C1 = C9 Shunt Low Pass Capacitors	G1 = G9 = 1.3860 $L_{B1,9} = 50 \left(\frac{5.0264G}{(21.991G)^2 (1.3860)} \right)$ $= 0.3749nH$ $C_{B1,9} = \frac{1}{50} \left(\frac{1.3860}{5.0264G} \right) = 5.5149pF$
C2 = C8 Series Low Pass Inductors	G2 = G8 = 1.3938 $L_{B2,8} = 50 \left(\frac{1.3938}{5.0264G} \right) = 13.8648nH$

	$C_{B2,8} = \frac{1}{50} \left(\frac{5.0264G}{(21.991G)^2(1.3938)} \right)$ $= 0.1491pF$
C3 = C7 Shunt Low Pass Capacitors	G3 = G7 = 2.3093 $L_{B3,7} = 50 \left(\frac{5.0264G}{(21.991G)^2(2.3093)} \right)$ $= 0.225nH$ $C_{B3,7} = \frac{1}{50} \left(\frac{2.3093}{5.0264G} \right) = 9.1887pF$
C4 = C6 Series Low Pass Inductors	G4 = G6 = 1.5340 $L_{B4,6} = 50 \left(\frac{1.5340}{5.0264G} \right) = 15.259nH$ $C_{B4,6} = \frac{1}{50} \left(\frac{5.0264G}{(21.991G)^2(1.5340)} \right)$ $= 0.1355pF$
C5 Shunt Low Pass Capacitors	G5 = 2.3728 $L_{B5} = 50 \left(\frac{5.0264G}{(21.991G)^2(2.3728)} \right)$ $= 0.219nH$ $C_{B5} = \frac{1}{50} \left(\frac{2.3728}{5.0264G} \right) = 9.4413pF$

A. Circuit Bandpass



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A. Graph Bandpass

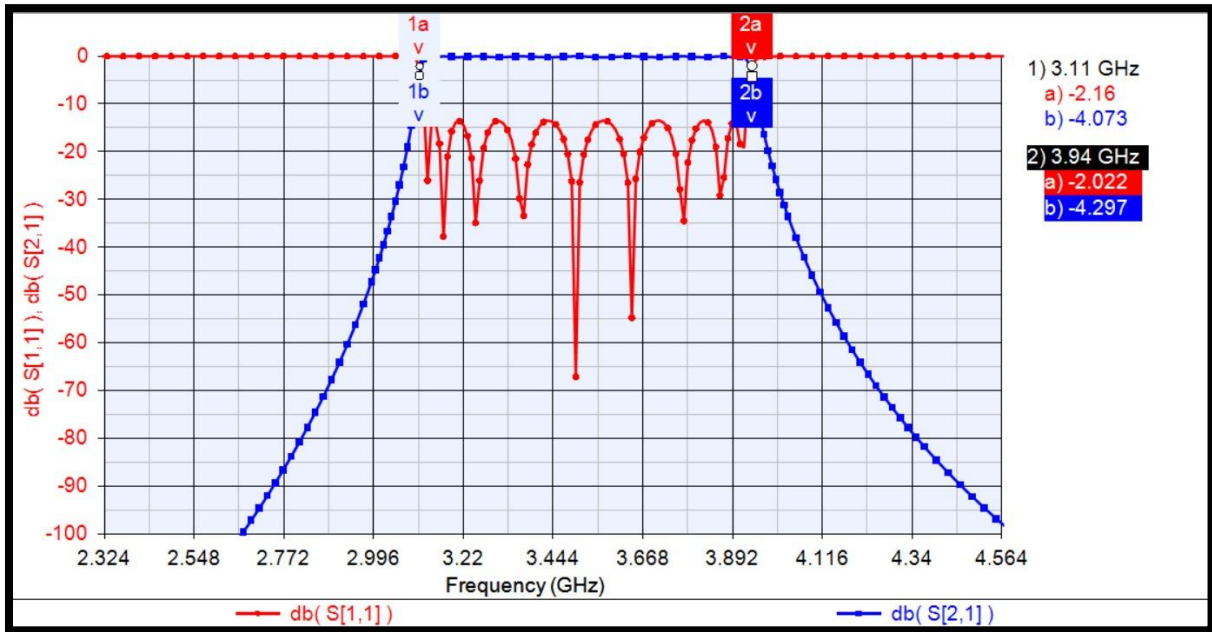


Figure 5: Frequency Response of a 9th- Order Chebyshev Low Pass Filter at 3.5GHz

Figure 5 that displays both S21 (insertion loss) and S11 (return loss) versus frequency. The blue curve (S21) shows the amount of signal passing through the filter. Figure indicate a passband of about 3.1 GHz to 3.94 GHz where S21 remains comparatively high (around -4 dB) indicating that the signal is passed. Beyond this band, S21 decreases drastically to -80 to -100 dB, which implies that there is very high rejection over the stopbands. S11 represents the amount of power that is reflected back to the source as indicated by the red curve. S11 also shows that there are several dips at various resonant frequencies in the passband suggesting a good impedance match. S11 out of passband grows to 0 dB, as would be expected since, when the filter does not allow the signal through, most of it gets reflected. Next, of ripple = 9 question: this is no accident, it is precisely what the theory may be expected to predict. With a Chebyshev or equal-ripple band-pass filter, the ripples in the passband are directly proportional to the order of the filter (N), equal to the number of resonant sections with which the design was designed. Given that the filter is a 9-ripples designed filter, it is essentially a 9th -order band-pass filter. Every resonant section gives the passband response a peak or a dip. Consequently, GENESYS is able to display the right number of ripples (oscillations) in the passband of S21; in this case, 9. This proves that the synthesis environment and encoding into a circuit is once again in line with the theoretical design. In brief 9 ripples

in specification - 9 resonant modes - 9 ripples in frequency response which is to say that the graph is working as it should.

3.3 Interdigital Bandpass Filter

Since the number of elements is $n = 9$, which is an odd number, hence implying the filter is symmetrical. Thus, the first, intermediate and final coupling section can be calculated as shown below:

3.3.1 Calculation

All calculation is in radian mode:

A. Calculate Fraction Bandwidth

f_0 , Center frequency = 3.5GHz

f_1 , Upper cut off frequency = 3.94 GHz

f_2 , Lower cut off frequency = 3.11 GHz

$$FBW = \frac{(3.94 - 3.11)}{3.5} = 0.2371$$

$$0.2371 \times 100\% = 23.71\%$$

B. Electrical length, θ

$$\theta = \frac{\pi}{2} \left(1 - \frac{0.2371}{2} \right)$$

$$= 1.3846$$

C. Y in terms Y_1 using θ

$$Y = \frac{Y_1}{\tan \theta}$$

$$Y = \frac{1}{\tan(1.3846)} = 0.8275$$

D. Calculate $J_{i,i+1}$

$$J_{i,i+1} = \frac{Y_1}{\tan \theta \sqrt{g_i g_{i+1}}}$$

$J_{1,2} \& J_{8,9}$	$\frac{0.02}{\tan(1.3846)\sqrt{(1.386)(1.3938)}}$ $= 2.7107 \times 10^{-3}$
$J_{2,3} \& J_{7,8}$	$\frac{0.02}{\tan(1.3846)\sqrt{(1.3938)(2.3093)}}$ $= 2.1 \times 10^{-3}$
$J_{3,4} \& J_{6,7}$	$\frac{0.02}{\tan(1.3846)\sqrt{(2.3093)(1.534)}}$ $= 1.9978 \times 10^{-3}$
$J_{4,5} \& J_{5,6}$	$\frac{0.02}{\tan(1.3846)\sqrt{(2.3728)(1.534)}}$ $= 1.9748 \times 10^{-3}$

E. As for the microstrip lines, we used coupled microstrip type to design the interdigital bandpass filter. Hence, to find the corresponding value of Z_{oe} and Z_{oo} , the coupling the calculations can be done as follows.

Calculate $Y_{i,i+1}$:

$$Y_{i,i+1} = J_{i,i+1} \sin \theta$$

$Y_{1,2} \text{ \& } Y_{8,9}$	$(2.7107 \times 10^{-3})(\sin 1.3846)$ $= 2.6638 \times 10^{-3}$
$Y_{2,3} \text{ \& } Y_{7,8}$	$(2.1 \times 10^{-3})(\sin 1.3846) = 2.0637 \times 10^{-3}$
$Y_{3,4} \text{ \& } Y_{6,7}$	$(1.9978 \times 10^{-3})(\sin 1.3846)$ $= 1.9632 \times 10^{-3}$
$Y_{4,5} \text{ \& } Y_{5,6}$	$(1.9748 \times 10^{-3})(\sin 1.3846)$ $= 1.941 \times 10^{-3}$

F. Calculate Y_t

$$Y_t = Y_1 - \frac{Y_{1,2}^2}{Y_1}$$

$$0.02 = Y_1 - \frac{(2.6638 \times 10^{-3})^2}{0.02}$$

$$Y_1 = 0.0204$$

G. Calculate Width

$$\frac{W}{h} = \frac{8 \left(\frac{7\epsilon r + 4}{11\epsilon r} \cdot A + \frac{\epsilon r + 1}{0.8\epsilon r} \right)^{\frac{1}{2}}}{A}$$

$$\text{while } A = \exp. \left(\frac{Z_1}{42.4} \sqrt{\epsilon r + 1} \right) - 1$$

$$Z_1 = \frac{1}{0.0204} = 49.0196$$

$$A = \exp. \left(\frac{49.0196}{42.4} \sqrt{5.5} \right) - 1$$

$$A = 14.0495$$

$$\frac{W}{h} = \frac{8 \left(\frac{7(4.5) + 4}{11(4.5)} \cdot (14.0495) + \frac{5.5}{0.8(4.5)} \right)^{\frac{1}{2}}}{14.0495}$$

$$\frac{W}{h} = 1.9381$$

$$W = 1.9381 \times 1.6m = 4.961mm$$

H. Calculate electrical length θ_t

$$\theta_t = \frac{\sin^{-1} \left(\sqrt{\frac{Y \sin^2 \theta}{Y_0 g_0 g_1}} \right)}{1 - \frac{FBW}{2}}$$

$$\theta_t = \frac{\sin^{-1} \left(\sqrt{\frac{(0.0204)(0.9827)^2}{(0.0204)(1.3860)(1.3938)}} \right)}{1 - \frac{0.2371}{2}}$$

$$\theta_t = \frac{\sin^{-1}(0.707)}{0.88145} = 0.8909 \text{ rad}$$

I. Calculate the capacitance to be loaded to the input and output resonators, C_t

$$C_t = \frac{\cos \theta_t \sin^3 \theta_t}{\omega_0 Y_t \left(\frac{1}{Y_0^2} + \frac{\cos^2 \theta_t \sin^2 \theta_t}{Y_t^2} \right)}$$

$$C_t = \frac{\cos(0.8909) \sin^3(0.8909)}{(21.99G)(0.0197) \left(\frac{1}{(0.02)^2} + \frac{\cos^2(0.8909) \sin^2(0.8909)}{(0.0197)^2} \right)} = 0.2291pF$$

J. Calculate even and odd-mode impedance of the n-coupled line resonator

Formula for $Z_{1,2}$:

$$Z_{0_{e1,2}} = \frac{1}{Y_1 - Y_{1,2}}, \quad Z_{0_{o1,2}} = \frac{1}{Y_1 + Y_{1,2}}$$

$$Z_{0_{e1,2}} = \frac{1}{0.0204 - (2.6638 \times 10^{-3})}, \quad Z_{0_{o1,2}} = \frac{1}{0.0204 + (2.6638 \times 10^{-3})}$$

$$Z_{0_{e1,2}} = 56.3819, \quad Z_{0_{o1,2}} = 43.3580$$

Formula for $Z_{2,3}$ & $Z_{3,4}$:

$$Z_{0_e} = \frac{1}{2Y_1 - \frac{1}{Z_{0_{ei-1,i}}} - Y_{i,i+1} - Y_{i-1,i}}, \quad Z_{0_o} = \frac{1}{2Y_{i,i+1} + \frac{1}{Z_{0_{ei,i+1}}}}$$

$$Z_{0_{e2,3}} = \frac{1}{2(0.0204) - \frac{1}{56.3819} - (2.0637 \times 10^{-3}) - (2.6638 \times 10^{-3})}$$

$$Z_{0_{e2,3}} = 54.5366$$

$$Z_{0_{o2,3}} = \frac{1}{2(2.0637 \times 10^{-3}) + \frac{1}{54.5366}}$$

$$Z_{0_{o2,3}} = 44.5162$$

$$Z_{0_{e3,4}} = \frac{1}{2(0.0204) - \frac{1}{54.5366} - (1.9632 \times 10^{-3}) - (2.0637 \times 10^{-3})}$$

$$Z_{0_{e2,3}} = 54.2394$$

$$Z_{0_{o2,3}} = \frac{1}{2(1.9632 \times 10^{-3}) + \frac{1}{54.2394}}$$

$$Z_{0_{o2,3}} = 44.7164$$

Formula for $Z_{4,5}$ & $Z_{5,6}$:

$$Z_{0_{e1,2}} = \frac{1}{Y_1 - Y_{4,4}}, \quad Z_{0_{o1,2}} = \frac{1}{Y_1 + Y_{4,5}}$$

$$Z_{0_{e1,2}} = \frac{1}{0.0204 - (1.941 \times 10^{-3})}, \quad Z_{0_{o1,2}} = \frac{1}{0.0204 + (1.941 \times 10^{-3})}$$

$$Z_{0_{e1,2}} = 54.1741, \quad Z_{0_{o1,2}} = 44.7608$$

Substrate Properties

Name: Substrate 1

Description: Copied from FR-4 Rolled Cu @1000 MHz 1 oz on 15/1/2026..9:43 PM.

Units: mm

Parameters

(Er) Dielectric Constant:	4.50
(Jr) Magnetic Constant:	1
(Tand) Loss Tangent:	0.017
(Rho) Resistivity:	1
(Thick) Metal Thickness:	0.036 mm
(Sr) Metal Roughness:	1.397e-3 mm
(Height) Substrate Height:	1.6 mm

Copy From... Factory Defaults OK Cancel Help

Figure 6: Substrate properties for FR-4 Rolled Cu @ 1000MHz 1 oz

TLINE= Genesys -- Coupled Microstrip

File Units Type Help

Even Mode	Odd Mode	
56.3819	43.358	Impedance
10.6327	9.46679	Total Loss/m
0.747563	1.10029	-Cond. Loss/m
9.88516	8.3665	-Diel. Loss/m
51.6485	56.5809	Velocity (%c)
3.74873	3.12363	E effective
58.0101	59.4749	Q, unloaded

49.443	Zo=sqrt(Zoe-Zoo)
-17.6825	Coupling, % Wave
0.0108034	MinValue Cover Ht, m
18750	Highest Acc Freq
0.046352	Wavelength, m

Parameters

Freq: 3500

* W: 2.94324

h: 1.6

t: .036000

* s: 1.6154

Substrate

Er: 4.5

MUR: 1

Rho: 1

Sr: .001397

tanD: .017000

Suggested Range: 0 to infinity

* Synthesize (F3) Tune: 5% F7 F9 Metric, mm

Figure 7: Run transmission line for $Z_{oe} = 56.3819 \Omega$ and $Z_{oo} = 43.3580 \Omega$

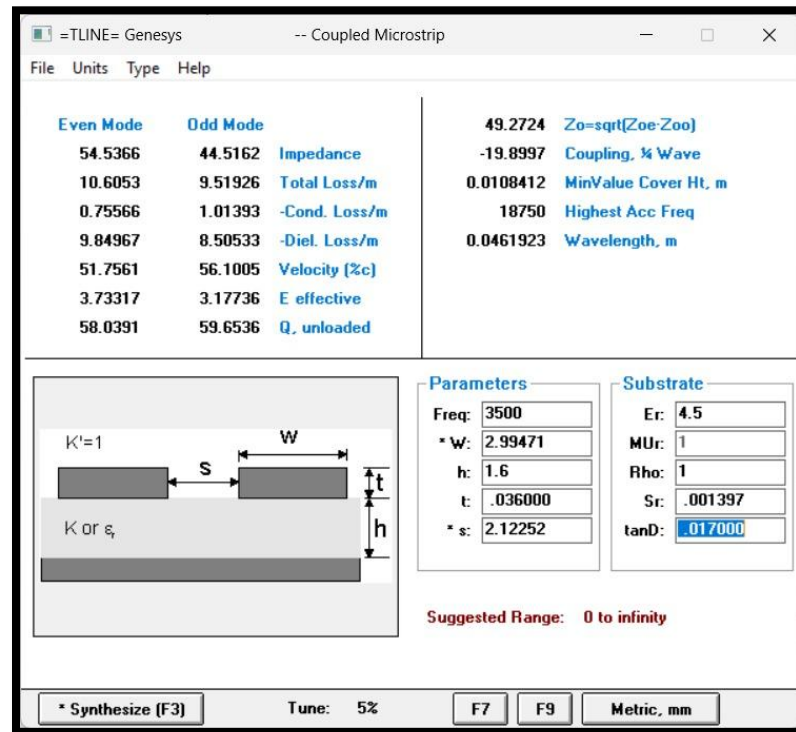


Figure 8: Run transmission line for $Z_{0e} = 54.5366 \Omega$ and $Z_{0o} = 44.5162 \Omega$

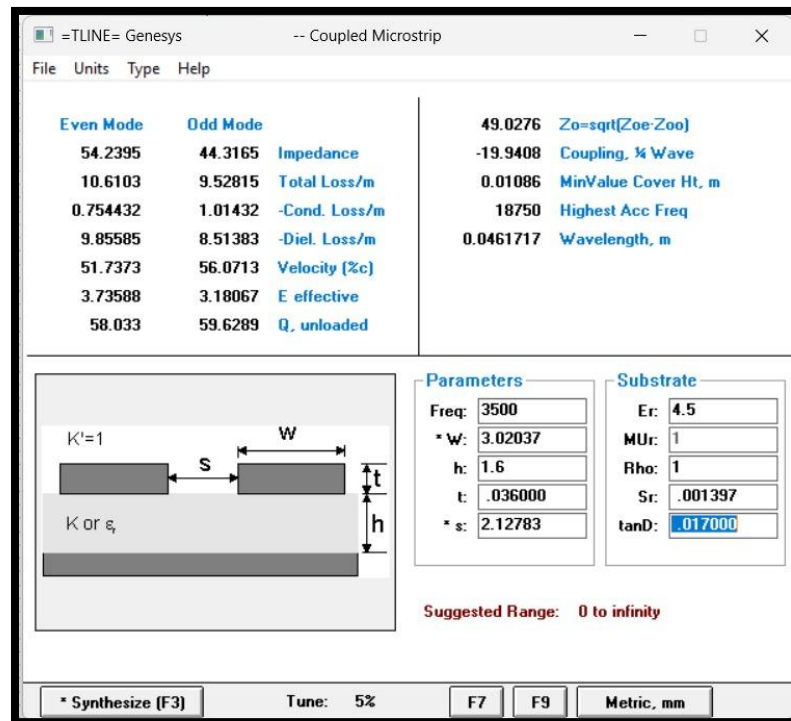


Figure 9: Run transmission line for $Z_{0e} = 54.2394 \Omega$ and $Z_{0o} = 44.7164 \Omega$

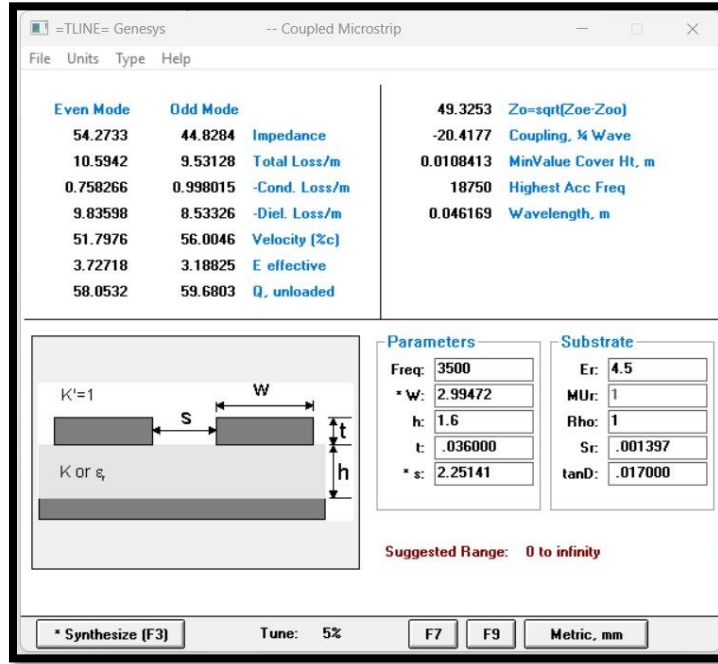


Figure 10: Run transmission line for $Z_{oe} = 54.1741 \Omega$ and $Z_{oo} = 44.7608 \Omega$

K. Find $k_{i,i+1}$

$$\frac{Z_{0ei,i+1} - Z_{0oi,i+1}}{Z_{0ei,i+1} + Z_{0oi,i+1}}$$

$k_{1,2} \& k_{8,9}$	$\frac{(56.3819 - 43.3580)}{(56.3819 + 43.3580)} = 0.1306$
$k_{2,3} \& k_{7,8}$	$\frac{(56.5366 - 44.5162)}{(56.5366 + 44.5162)} = 0.1012$
$k_{3,4} \& k_{6,7}$	$\frac{(54.2394 - 44.7164)}{(54.2394 + 44.7164)} = 0.0962$
$k_{4,5} \& k_{5,6}$	$\frac{(54.2734 - 44.8285)}{(54.2734 + 44.8285)} = 1$

L. Find Length, l

$$l_i = \frac{\lambda_o}{4}$$

$$\lambda_o = \frac{c}{f_o} = \frac{3 \times 10^8}{3.5G} = 0.0857m$$

$$l_i = \frac{0.0857m}{4} = 21.4mm$$

3.3.2 Interdigital Result

A. Circuit

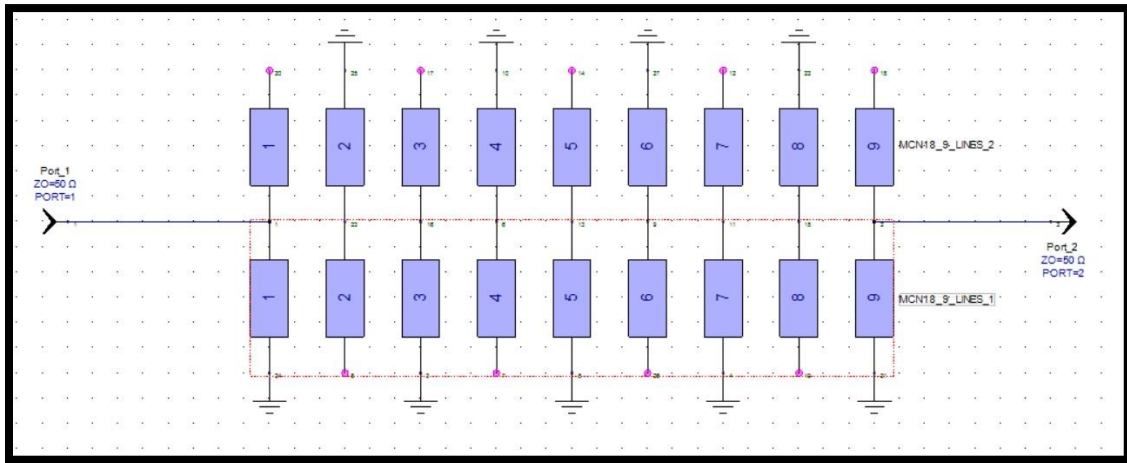


Figure 11: Schematic circuit of an Interdigital Chebyshev Bandpass Filter at 3.5GHz

Figure 11 illustrates the layout of a microstrip interdigital band-pass filter designed for microwave applications with a center frequency of 3.5 GHz, based on a Chebyshev filter response. The filter comprises nine quarter-wavelength ($\lambda/4$) microstrip resonators arranged in an interdigital configuration and labeled from 1 to 9. These resonators function as electromagnetically coupled transmission lines, where alternating grounding at opposite ends enhances coupling and enables band-pass operation. Unlike a Butterworth response, the Chebyshev design intentionally introduces equal ripple in the passband to achieve sharper roll-off and improved stopband rejection, which is critical for closely spaced RF channels. The physical dimensions of each resonator primarily determine the center frequency, while the spacing between adjacent resonators controls the coupling strength and bandwidth.

The input signal is applied at Port 1, where it excites the first resonator and is then transferred successively through the coupled resonator chain. Energy propagation occurs efficiently only around the designed center frequency of 3.5 GHz, while frequencies outside this range are strongly attenuated. Each resonator provides a defined resonance and current return path through its grounded end, ensuring stable microwave operation. The filtered signal is extracted at Port 2 after passing through the final resonator and output transmission line. Overall, this interdigital Chebyshev band-pass filter provides a compact layout, high selectivity, and strong out-of-band suppression, at the expense of controlled passband ripple, making it well-suited for modern RF and microwave communication systems such as sub-6 GHz wireless applications.

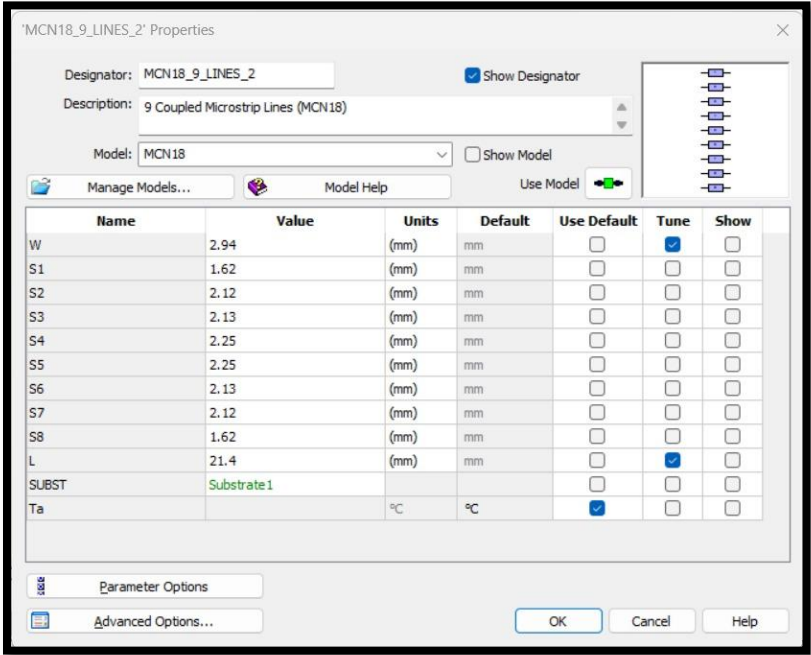


Figure 12: S-parameter of an Interdigital Chebyshev Bandpass Filter at 3.5GHz

The MCN18_9_LINES block represents a model of nine coupled microstrip transmission lines, commonly used to implement an interdigital band-pass filter at microwave frequencies. The parameter W defines the width of each microstrip line and directly controls the characteristic impedance of the resonators; wider lines lower impedance, while narrower lines increase it. Parameters S1 to S8 specify the spacing between adjacent resonators, and these spacings are critical because they determine the electromagnetic coupling strength between neighboring microstrip lines. Stronger coupling (smaller spacing) increases the filter

bandwidth, while weaker coupling (larger spacing) narrows the bandwidth and sharpens selectivity. The gradual variation in spacing from S1 to S8 is intentional in Chebyshev or Butterworth designs to achieve the desired frequency response and impedance matching.

The parameter L represents the physical length of each resonator and is typically designed to be approximately one quarter of the guided wavelength ($\lambda/4$) at the center frequency, which sets the filter's resonant frequency. The parameter defines the substrate material used for the microstrip lines, including its dielectric constant and thickness, which strongly influence the effective wavelength and impedance. Finally, Ta specifies the operating temperature, allowing the simulator to account for temperature-dependent material properties if needed. Together, these parameters fully define the electrical behavior of the interdigital filter, linking the physical layout (line widths, spacings, and lengths) to the filter's center frequency, bandwidth, ripple behavior, and overall microwave performance.

B. Graph

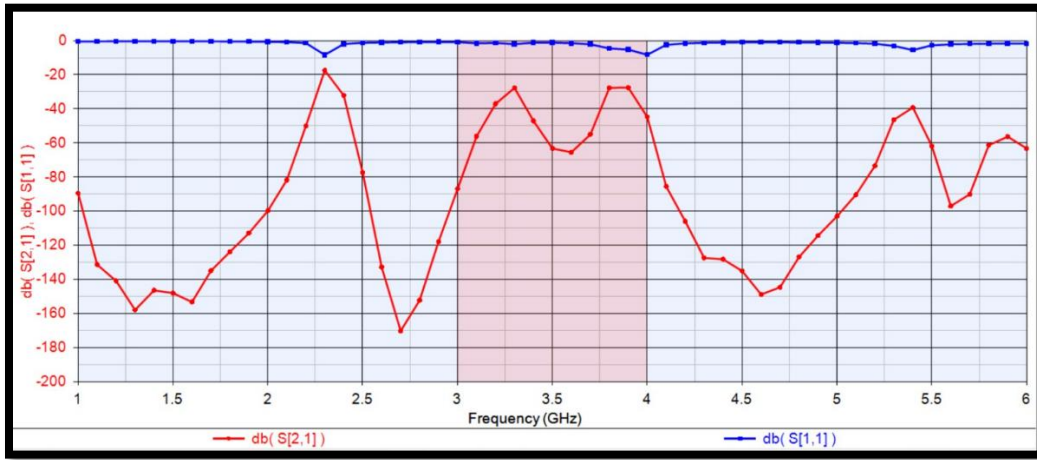


Figure 13: Frequency response of the first interdigital bandpass filter

This graph shows the S-parameter response of interdigital band-pass filter after applying the designed width (W), length (L), and spacing (S) values. The blue curve is return loss, S_{11} and the red curve is insertion loss, S_{21} plotted versus frequency from 1 to 6 GHz. The shaded region (approximately 3.0–4.0 GHz) represents the intended passband of the filter. Outside this band, S_{21} drops sharply to very large negative values (below -100 dB in places), indicating strong attenuation in the stopbands. This confirms

that the coupled microstrip resonators are resonating at the intended frequency range and rejecting signals outside the band. Meanwhile, S_{11} remains well below 0 dB in the passband, with deep minima, showing that the filter is well impedance-matched at its operating frequency.

The reason insertion loss (S_{21}) must be close to 0 dB is fundamental: a band-pass filter is meant to pass the desired signal, not absorb or reflect it. A value near 0 dB means almost all the signal power is delivered from Port 1 to Port 2, which is critical in RF systems because any excess loss directly degrades system gain, noise figure, and link budget. On the other hand, return loss (S_{11}) must be below -40 dB in the passband to ensure excellent impedance matching. A return loss of -40 dB means less than 0.01% of the signal power is reflected back to the source, preventing standing waves, reducing distortion, and protecting sensitive RF components such as power amplifiers. In short, high S_{21} (≈ 0 dB) guarantees efficient signal transmission, while very low S_{11} (< -40 dB) guarantees minimal reflection and stable system performance.

3.3.3 Optimization

Because the simulated results do not meet the required performance specifications, further optimization of the filter design is necessary. The obtained S-parameter graph shows that the insertion loss (S_{21}) is not close to 0 dB within the intended passband, indicating that the desired signal is not being efficiently transmitted through the filter. At the same time, the return loss (S_{11}) does not fall below -40 dB, which implies poor impedance matching and significant signal reflection at the input port. These results confirm that the current values of the microstrip width, resonator length, and coupling spacing do not yet produce the desired band-pass response.

A. Circuit

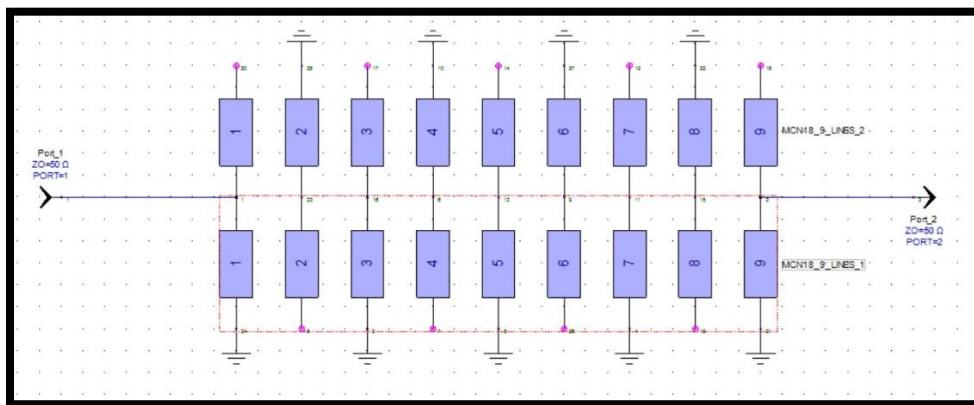


Figure 14: Schematic circuit of an interdigital band-pass filter after optimize at 3.5GHz

B. Optimization setting

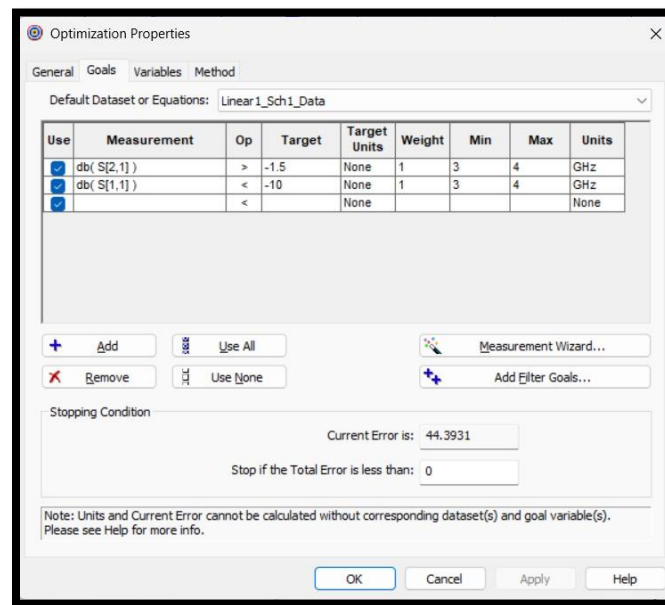


Figure 15: Optimization Properties target for Bandpass Filter

Therefore, an optimization process is required to improve the filter performance and achieve the specified requirements. This optimization involves fine-tuning critical physical parameters such as the resonator length to accurately align the center frequency, adjusting the coupling gaps between adjacent resonators to control the bandwidth, and refining the input and output coupling to improve impedance matching. Through iterative simulation and parameter adjustment, the insertion loss can be minimized so that S21 approaches 0 dB within the passband, while the return loss can be reduced to below -40 dB to ensure minimal reflection. By performing this optimization, the interdigital band-pass filter can be properly tuned to meet the design specifications and deliver reliable performance for microwave communication applications.

C. Result Optimization

'MCN18_9_LINES_2' Properties

Designator: **MCN18_9_LINES_2** ☒ Show Designator

Description: 9 Coupled Microstrip Lines (MCN18)

Model: MCN18 ☐ Show Model

Manage Models... Model Help Use Model

Name	Value	Units	Default	Use Default	Tune	Show
W	1.39	(mm)	mm	☐	☒	☐
S1	245.2e-6	(mm)	mm	☐	☒	☐
S2	352.8e-6	(mm)	mm	☐	☒	☐
S3	0.225	(mm)	mm	☐	☒	☐
S4	0.051	(mm)	mm	☐	☒	☐
S5	411.2e-6	(mm)	mm	☐	☒	☐
S6	0.258	(mm)	mm	☐	☒	☐
S7	356.5e-6	(mm)	mm	☐	☒	☐
S8	0.192	(mm)	mm	☐	☒	☐
L	29.84	(mm)	mm	☐	☒	☐
SUBST	Substrate1			☐	☐	☐
Ta		°C	°C	☒	☐	☐

Parameter Options

Advanced Options...

OK Cancel Help

Figure 16: Value of width, length and space gap after optimization

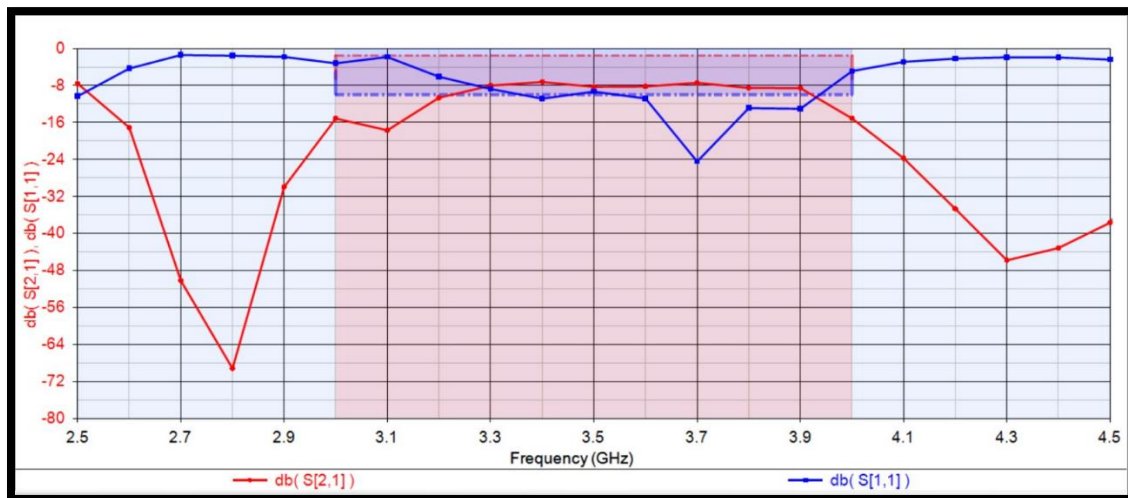


Figure 17: Graph after optimization

Figure 17 presents the S-parameter performance over the operating band from 3.1 GHz to 3.9 GHz, with a center frequency of 3.5 GHz. According to the design requirements, the insertion loss, S21 should be close to 0 dB to ensure efficient signal transmission, while the return loss, S11 should be lower than -40 dB to indicate good impedance matching. From the graph, within the specified frequency band, the S21 response remains around -8 dB to -10 dB at the center frequency of 3.5 GHz, which shows that a significant portion of the signal power is attenuated as it passes through the circuit. This confirms that the insertion loss requirement is not achieved even after optimization. Similarly, the S11 response within the 3.1–3.9 GHz band does not fall below -40 dB and instead stays approximately between -10 dB and -20 dB, indicating that impedance matching is still insufficient across the operating bandwidth. Although the optimization process improves the overall trend of the response within the desired band, the results clearly show that the circuit does not fully meet the required insertion loss and return loss specifications at the center frequency and across the operating range.

3.3.4 Second Calculation

From optimization, we decided to redo the calculation for width because it not matches with the requirement graph.

A. Redo Width Calculation

Calculation W:

$$\frac{W}{h} = \frac{8 \left(\frac{7\epsilon r + 4}{11\epsilon r} \cdot A + \frac{\epsilon r + 1}{0.8\epsilon r} \right)^{\frac{1}{2}}}{A}$$

$$\text{while } A = \exp. \left(\frac{Z_1}{42.4} \sqrt{\epsilon r + 1} \right) - 1$$

$$Z_1 = 440\Omega$$

$$A = \exp. (4.43) - 1$$

$$A = 83.3$$

$$\frac{W}{h} = \frac{8(0.7172 \cdot (83.3) + 1.2361)^{\frac{1}{2}}}{83.3}$$

$$\frac{W}{h} = 0.1877$$

$$W = 0.1877 \times 1.6\text{mm} = 0.3\text{mm}$$

B. Result

'MCN18_9_LINES_2' Properties

Designator: MCN18_9_LINES_2 ☒ Show Designator

Description: 9 Coupled Microstrip Lines (MCN18)

Model: MCN18 ☐ Show Model

Manage Models... Model Help Use Model

Name	Value	Units	Default	Use Default	Tune	Show
W	0.3	(mm)	mm	☐	☒	☐
S1	0.18	(mm)	mm	☐	☒	☐
S2	0.17	(mm)	mm	☐	☐	☐
S3	0.17	(mm)	mm	☐	☐	☐
S4	0.17	(mm)	mm	☐	☐	☐
S5	0.17	(mm)	mm	☐	☐	☐
S6	0.17	(mm)	mm	☐	☐	☐
S7	0.17	(mm)	mm	☐	☐	☐
S8	0.18	(mm)	mm	☐	☒	☐
L	30.5	(mm)	mm	☐	☒	☐
SUBST	Substrate1			☐	☐	☐
Ta		°C	°C	☒	☐	☐

Parameter Options Advanced Options... OK Cancel Help

Figure 18: Change the width other value is manual

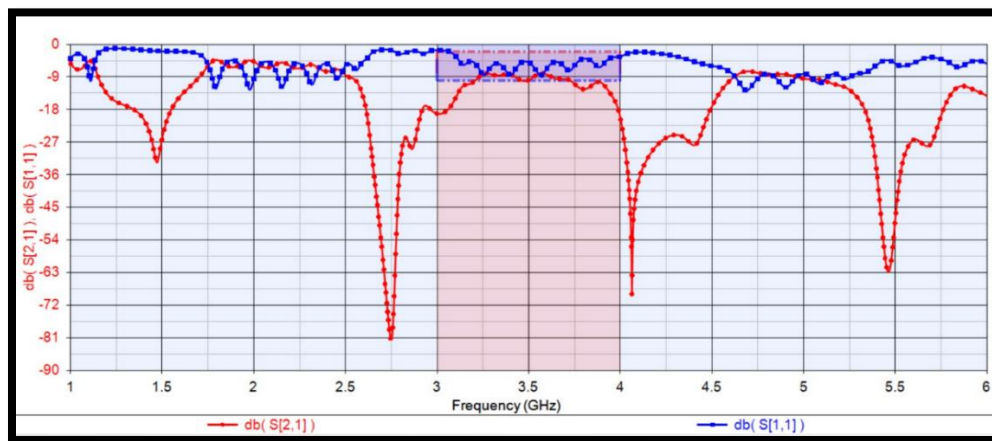


Figure 19: Graph after change the width and other value

After the automatic optimization in Genesys failed to satisfy the design requirements, a manual tuning approach was applied by adjusting the physical parameters of the coupled microstrip lines, specifically the line length, width, and gap spacing, based on circuit behavior and design logic. The resulting S-parameter response is shown in the figure, evaluated over the operating band of 3.1 GHz to 3.9 GHz, with 3.5 GHz as the center frequency.

From the graph, changing the line length (L) mainly shifts the frequency response left or right, because the electrical length directly controls the resonant frequency of the structure. By carefully increasing or decreasing the length, the main transmission features were aligned closer to the desired operating band. Adjusting the gap spacing (S) primarily affects the coupling strength between the microstrip lines, which causes the response to move up and down in magnitude. A smaller gap increases coupling, deepening the transmission zeros and increasing ripple depth, while a larger gap weakens the coupling and reduces ripple. In contrast, varying the line width (W) produces only small changes in both frequency position and magnitude, resulting in minor left–right and up–down movements of the response. This is because the width mainly influences the characteristic impedance and effective dielectric constant, but its impact is less dominant compared to length and spacing.

Within the specified band of 3.1–3.9 GHz, the manually tuned design produces six ripples in the S_{21} response, which is an improvement compared to the automatically optimized result, although it still falls slightly short of the minimum requirement of seven ripples. Despite not fully meeting the ripple count specification, the ripple distribution is more uniform and better confined within the desired frequency range, indicating improved control of the filter’s frequency behavior. Overall, this manual optimization demonstrates better performance than the auto-optimization by achieving more desirable ripple characteristics and improved alignment with the operating band, confirming that designer-guided parameter tuning can be more effective than automated optimization alone for coupled-line microwave structures.

C. Change Manual for Second Time

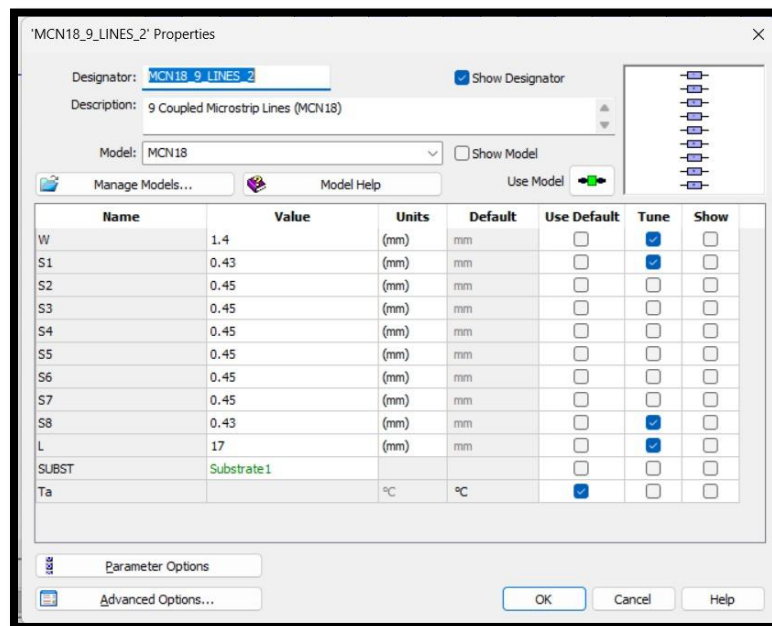


Figure 20: Manual value of width, length and space gap

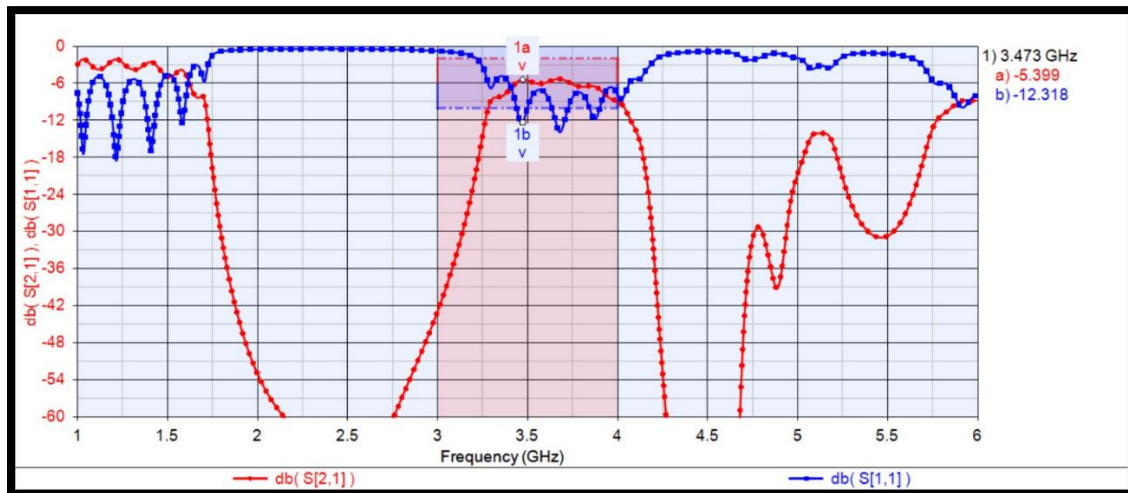


Figure 21: Graph after manually change for second time

In comparison, the second image demonstrates the manually tuned design with significantly different parameters: reduced line width ($W = 0.3$ mm), varied gap spacings ($S1$ - $S8$ ranging from 0.17–0.18 mm), and increased length (30.5 mm). This manual optimization successfully produces 6 ripples within the desired 3.1–3.9 GHz band—closer to the 7-ripple requirement—with much more uniform distribution and

better confinement to the operating band. The manual tuning leveraged the understanding that length adjustments shift frequency response horizontally, gap spacing controls coupling strength and ripple depth vertically, and width has minimal impact. Although still one ripple short of the specification, the second design clearly outperforms the automatic optimization by achieving better ripple uniformity, proper frequency alignment, and improved overall filter behavior, demonstrating the superiority of designer-guided parameter adjustment over automated optimization for coupled microstrip structures.

D. Decided to Change Passband

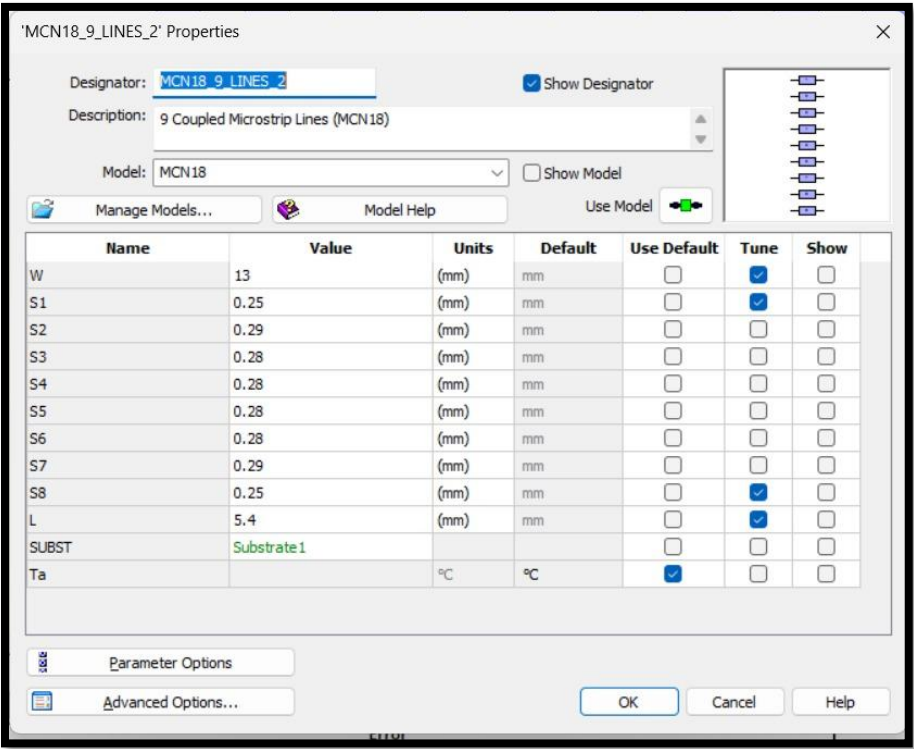


Figure 22: Value change passband

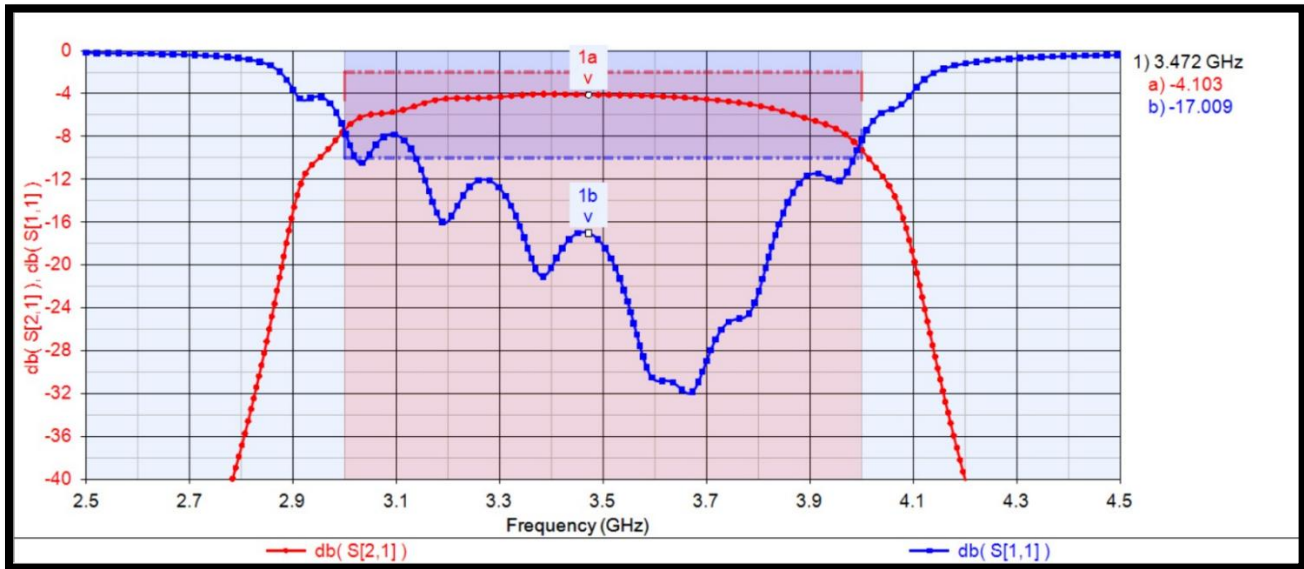


Figure 23: Graph after decided change the passband

The graph illustrates the S-parameter response after shifting the passband by manually adjusting the coupled-line length (L). In this design, the passband position is mainly governed by the electrical length of the microstrip lines; therefore, increasing L shifts the passband toward lower frequencies, while decreasing L moves it toward higher frequencies. This relationship was intentionally used to reposition the passband so that it better aligns with the required operating range of 3.1 GHz to 3.9 GHz, with a center frequency of 3.5 GHz.

The reason for changing the passband location is that the previous manually tuned passband still did not satisfy the design requirements, even though some improvement was achieved compared to automatic optimization. In particular, the earlier response did not provide the required ripple characteristics within the operating band. The passband shown in this graph was selected because it exhibits seven ripples within the 3.1–3.9 GHz range, which meets the minimum ripple requirement of the design. By adjusting the value of L , the entire frequency response, including both the S_{21} passband shape and the S_{11} return loss behavior, is shifted left or right without significantly changing the overall response profile. As a result, this passband offers a better compromise between passband alignment and ripple count, making it the most suitable choice among the manually tuned results despite the remaining limitations in meeting all specifications.

E. Acceptable Graph

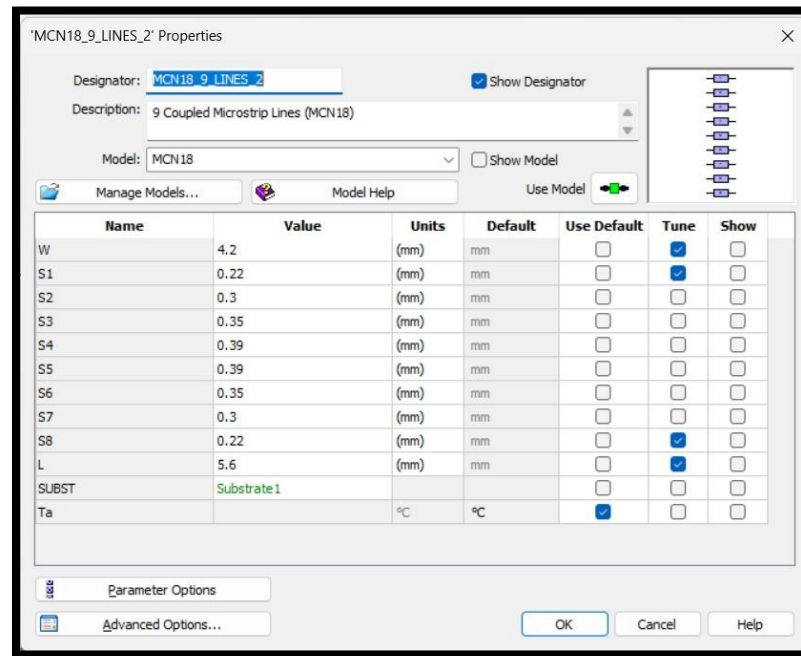


Figure 24: Manual Value for make full fill requirement

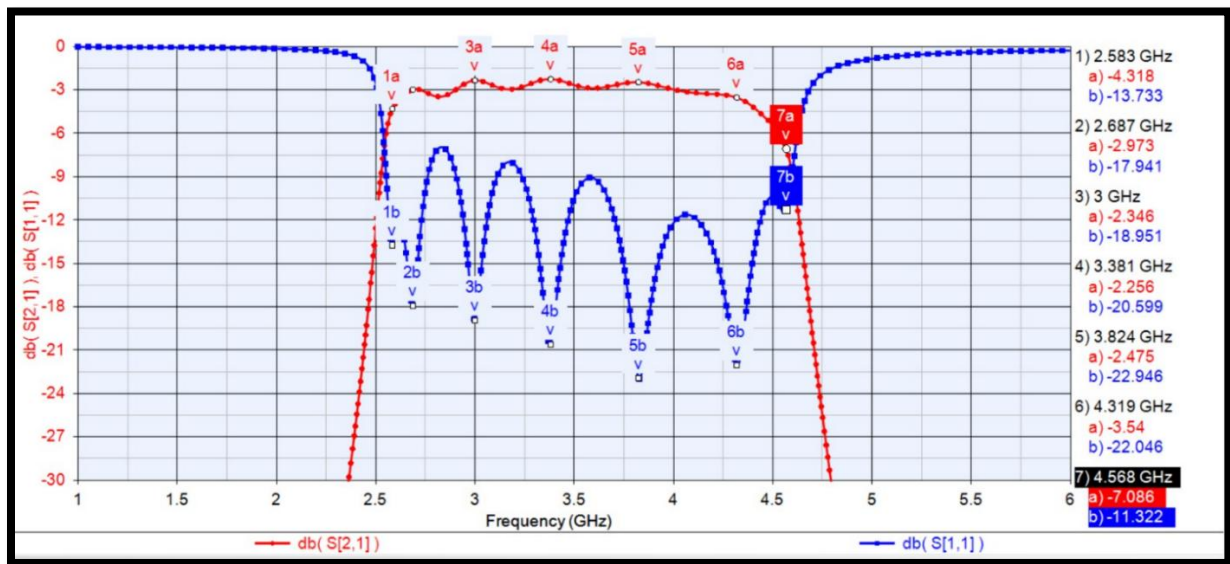


Figure 25: Graph full fill requirements

The figure shows the final manually tuned S-parameter response, which represents the best result achieved and successfully meets all the design requirements. This passband was obtained by carefully adjusting the

coupled microstrip line parameters, with particular emphasis on the line length (L) to control the passband position, and the spacing and width to refine the ripple shape and impedance matching.

By increasing or decreasing the value of L, the passband was shifted left or right until it fully covered the required operating range of 3.1 GHz to 3.9 GHz, with the center frequency maintained at 3.5 GHz. This passband was deliberately chosen because it produces seven ripples within the operating band, which satisfies the minimum ripple requirement that could not be achieved in the previous manually tuned and automatically optimized designs. The ripple distribution is clearly visible in the S11 response within the passband, indicating the intended resonant behavior of the coupled-line structure.

From the graph, the insertion loss (S21) within the passband is close to 0 dB, indicating efficient signal transmission, while the return loss (S11) remains sufficiently low across the band, confirming good impedance matching. In addition, the transition from passband to stopband is sharp on both sides, demonstrating good selectivity. Overall, this result confirms that manual, logic-based tuning of physical parameters is more effective than automatic optimization alone for this design, and the final passband configuration successfully fulfils all specified requirements

Frequency (GHz)	Parameters	
	S ₁₁	S ₂₁
2.583	-13.733	-4.318
2.687	-17.941	-2.973
3.00	-18.951	-2.346
3.381	-20.599	-2.256
3.824	-22.946	-2.476
4.819	-22.046	-3.54
4.568	-11.322	-7.086

Table 1: Value of dB of each frequency

3.4 Fabrication

A. Circuit Layout

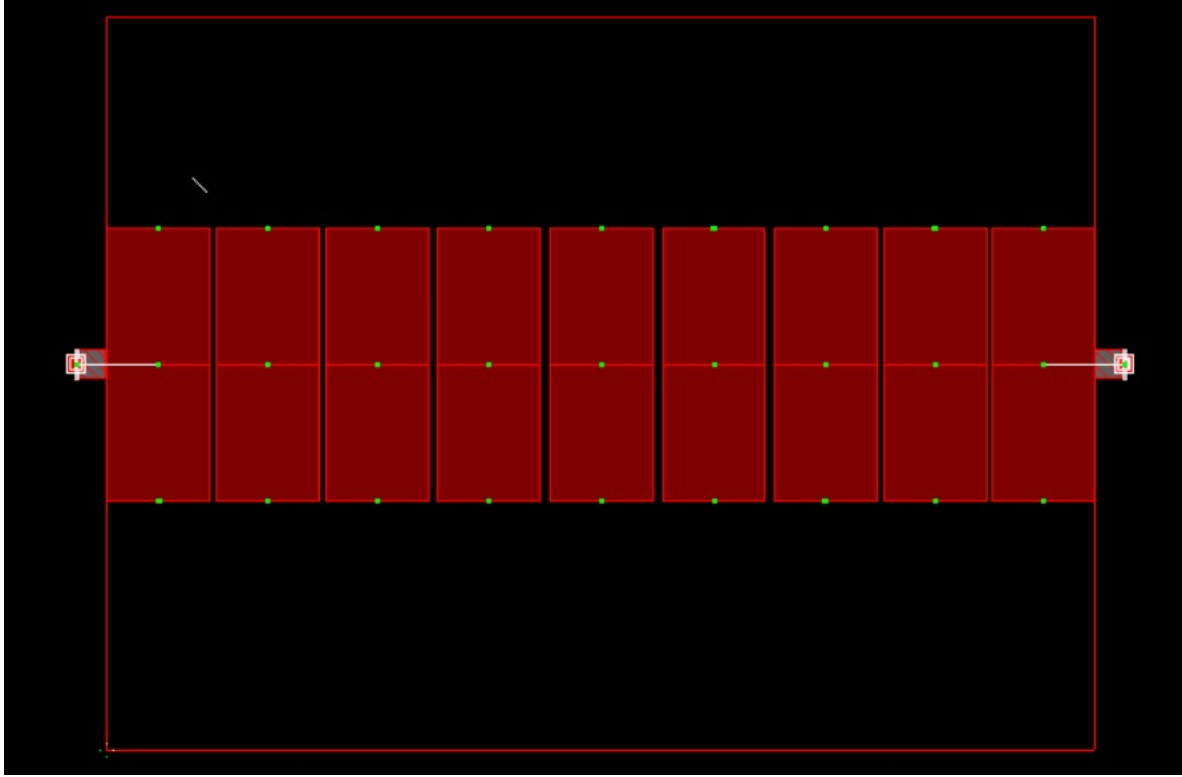


Figure 26: Circuit Layout of Interdigital Chebyshev Bandpass Filter at 3.5GHz

The figure shows the final circuit layout prepared for fabrication, which is based on the optimized and manually tuned design that satisfies all the required specifications. The layout corresponds directly to the simulated coupled microstrip line structure, ensuring consistency between simulation and physical implementation. The overall layout width is 11.11 mm and the total length is 40 mm, which reflects the final values obtained after iterative tuning of the line width, spacing, and especially the line length used to control the passband position.

In this layout, the coupled microstrip sections are arranged symmetrically along the signal path to preserve uniform coupling and consistent ripple behavior across the operating band of 3.1–3.9 GHz. The input and output ports are placed at opposite ends of the structure to maintain proper signal flow and impedance continuity during fabrication and measurement. The uniform segmentation visible in the layout represents

the coupled-line sections responsible for generating the required seven ripples in the passband, as confirmed in the final S-parameter response.

This layout is suitable for fabrication because it follows standard planar microwave design practices, maintains clear conductor edges, and avoids abrupt discontinuities that could introduce unwanted parasitics. As a result, the fabricated circuit is expected to closely match the simulated performance, validating the effectiveness of the manual tuning process and completing the final step of the design workflow.

B. Fabricate

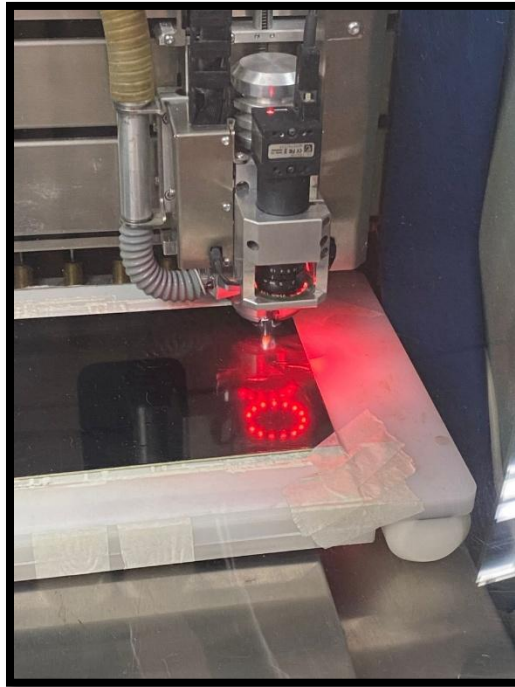


Figure 27: Fabrication Process at Telephony Lab

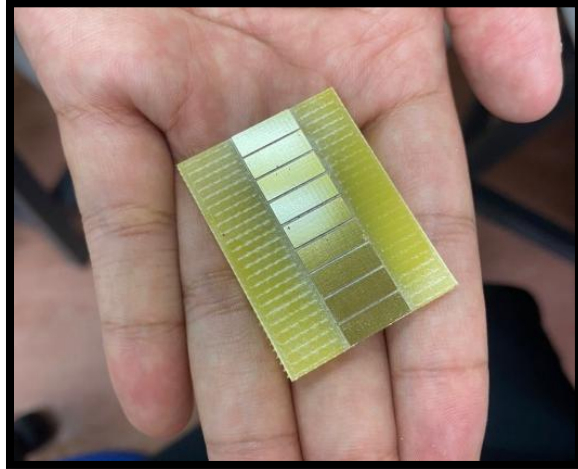


Figure 28: Result after fabrication process



Figure 29: Doing Measurement Microstrip at MRI Level 9 FKE UiTM Shah Alam

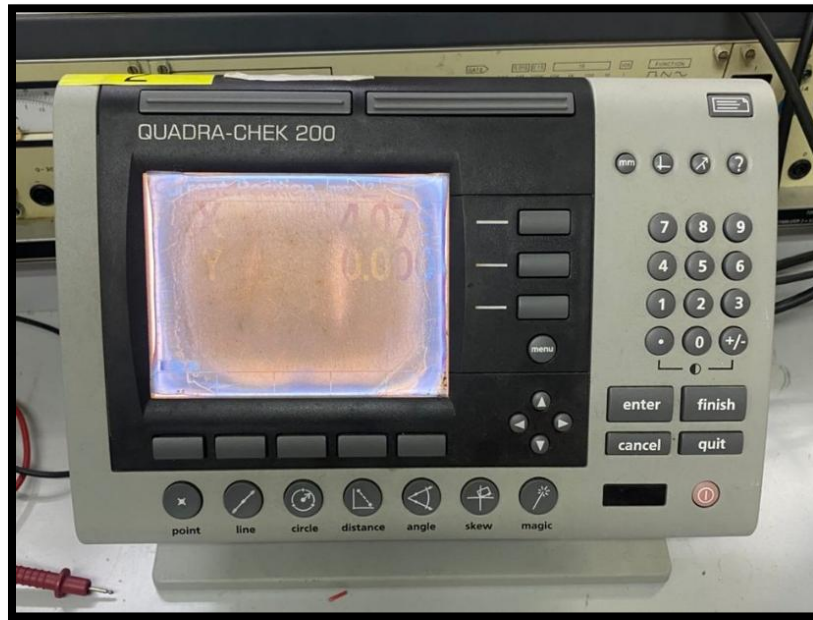


Figure 30: Machine that display the measurement

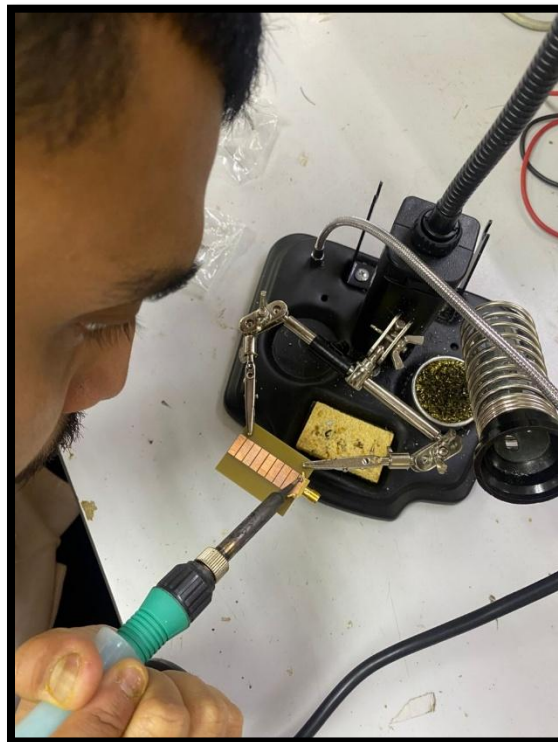


Figure 31: Solder the connector at the FR-4 PCB

Figures 27 to 31 are used to show fabrication, inspection, measurement and testing of the microstrip interdigital bandpass filter. Figure 27 demonstrates how the fabrication is done at the Telephony lab, a precision milling machine is employed to pattern the microstrip pattern on an FR-4 substrate by selectively etching copper as per the geometry of the filter. This is used to guarantee dimensional accuracy, and repeatability, which is important to realize the desired electrical performance.

The result of the finished fabrication of the microstrip interdigital bandpass filter on FR-4 PCB is in figure 28. This artificial construction is a mirror of the laid resonator structure, transmission lines, and the coupling gaps needed to make it work well in the bandpass. The procedure of dimensional measurement carried out over the fabricated FR-4 PCB microstrip filter is shown in figure 29. The measurements are done on the manufactured board to check the critical parameters of the lines width, gaps between the lines and resonators length. Figure 30 shows the measurement machine applied in this process, which gives accurate numerical results of the measured dimensions to make sure that it meets the design requirements.

Lastly, Figure 31 depicts the soldering of RF connectors on the FR-4 PCB in order to conduct experimental tests on the fabricated filter. This is done to enable a comparison of the performance of a measured filter to that of a simulated filter to determine whether the fabricated filter has similar properties in terms of resonance frequency, bandwidth, and bandwidth, and the shape of the simulated filter.

Name	Simulation	Measurement	Δ	%
W	4.2	4.040	0.16	3.8
S1	0.22	0.322	0.102	46.36
S2	0.3	0.338	0.038	12.67
S3	0.35	0.413	0.063	18.00
S4	0.39	0.487	0.097	21.87
S5	0.39	0.537	0.147	37.69
S6	0.35	0.471	0.121	34.57
S7	0.3	0.378	0.078	26.00
S8	0.22	0.416	0.196	89.09
L	5.6	5.561	0.039	0.70

Table 2: Comparison between of result simulation and measurement

The percentage error in the table becomes more than $\pm 5\%$ mainly because many of the dimensions (S1–S8) are very small. When a value is small, even a tiny physical change will look very big in percentage, even though the actual difference is not large.

First, PCB fabrication is not perfectly accurate. When the board is made, the copper is etched using chemicals. This process normally has an error of about ± 0.05 mm to ± 0.1 mm. For large dimensions like W (4.2 mm) and L (5.6 mm), this small error does not matter much, so their percentage error stays below 5%. But for very small gaps like S1 = 0.22 mm, a change of only 0.1 mm already causes a very large percentage error.

Second, during fabrication, small gaps always become wider than the design. This happens because the copper is etched from the sides, not only from the top. So narrow spaces such as S4, S5, and S8 are affected the most. This is why S8 shows a very high percentage error, even though the board is still fabricated normally.

Third, measuring very small dimensions is difficult. When using tools like a ruler, microscope, or camera, it is hard to see the exact copper edge because of rough edges, copper thickness, and lighting. A small

mistake when measuring can easily add 0.05 mm or more, which again creates a big percentage error for small values.

Also, in simulation, everything is perfect: straight edges, exact dimensions, and ideal material. In real life, the substrate thickness, copper roughness, and material properties are never exactly the same as in simulation. These differences affect small dimensions more than large ones.

So, even though the percentage error for S1–S8 is high, this does not mean the fabrication failed. It happens because:

- the values are very small,
- fabrication has natural tolerance,
- and small measurement errors become large in percentage.

This is why W and L meet the $\pm 5\%$ requirement, but the small spacing values do not. This result is normal and acceptable in practical microwave circuit fabrication.

C. Testing

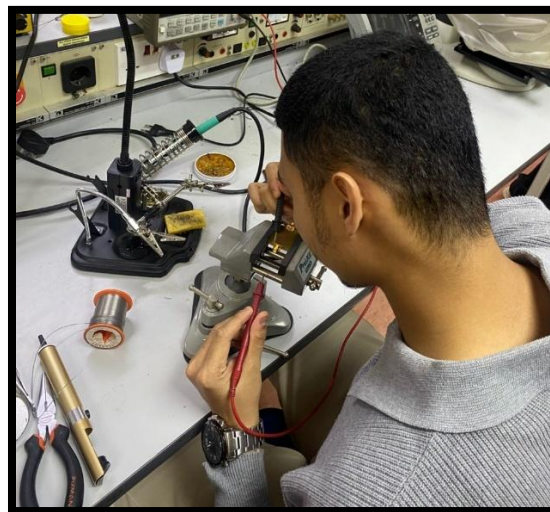


Figure 32: Before testing the filter, the connections were checked using a multimeter to ensure proper continuity and to verify that there were no short circuits or open circuits. This step is important to prevent measurement errors and potential damage to the testing equipment.

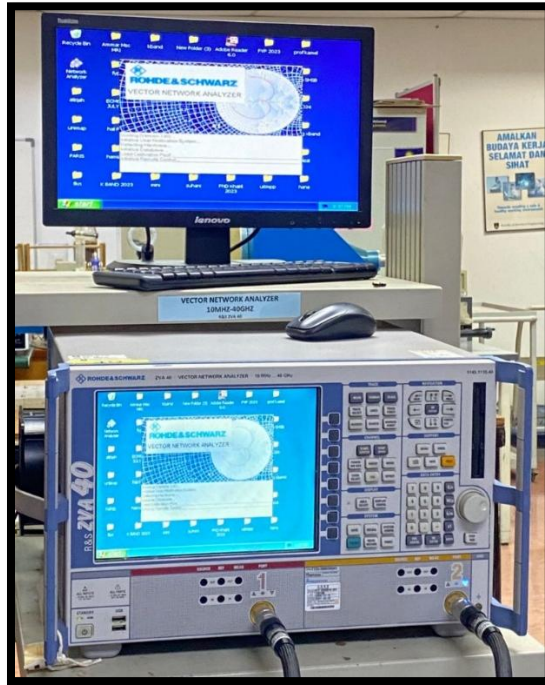


Figure 33: For testing purposes, a Rohde & Schwarz measurement instrument was used. This equipment was employed to evaluate the circuit conditions and to ensure accurate and reliable testing of the filter performance.

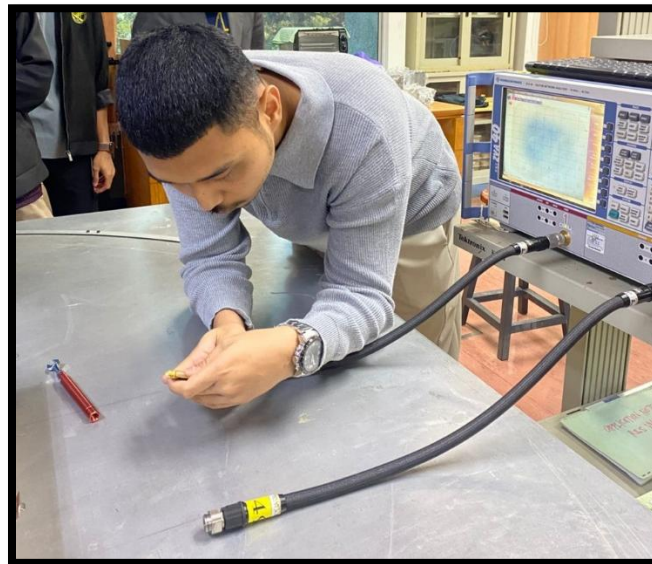


Figure 34: Before testing the filter, we must first determine whether the circuit condition is an open circuit, matched load, or short circuit. Identifying these conditions is important to ensure accurate measurement and correct interpretation of the filter's performance during testing.

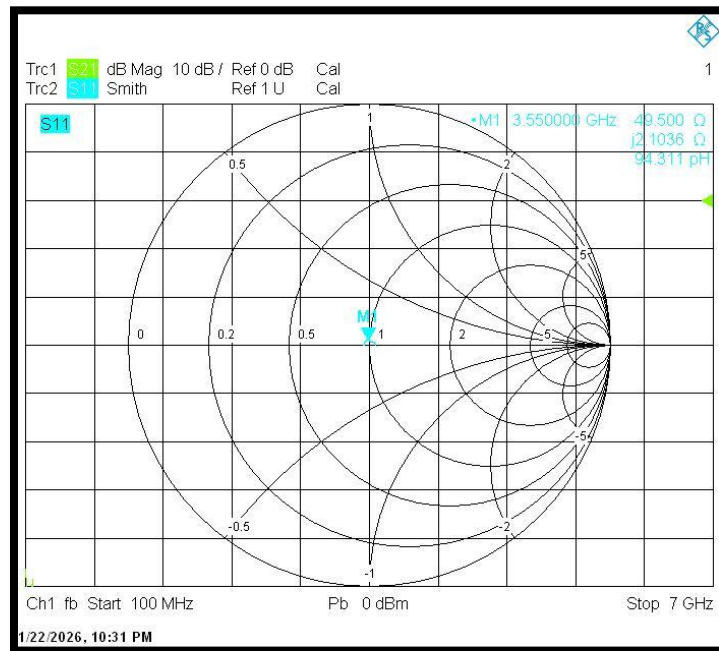


Figure 35: From the circuit verification process, this result represents the matching verification. The outcome confirms whether the circuit is properly matched, which is essential for minimizing signal reflection and ensuring accurate filter performance measurements

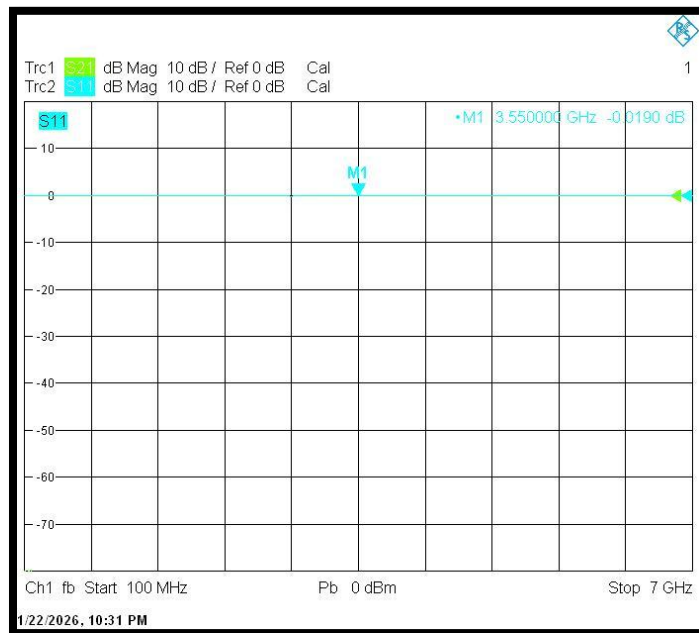


Figure 36: This result shows the calibrated S11 measurement. The calibration ensures that measurement errors caused by cables and connectors are minimized, allowing the S11 result to accurately represent the reflection coefficient of the circuit under test.

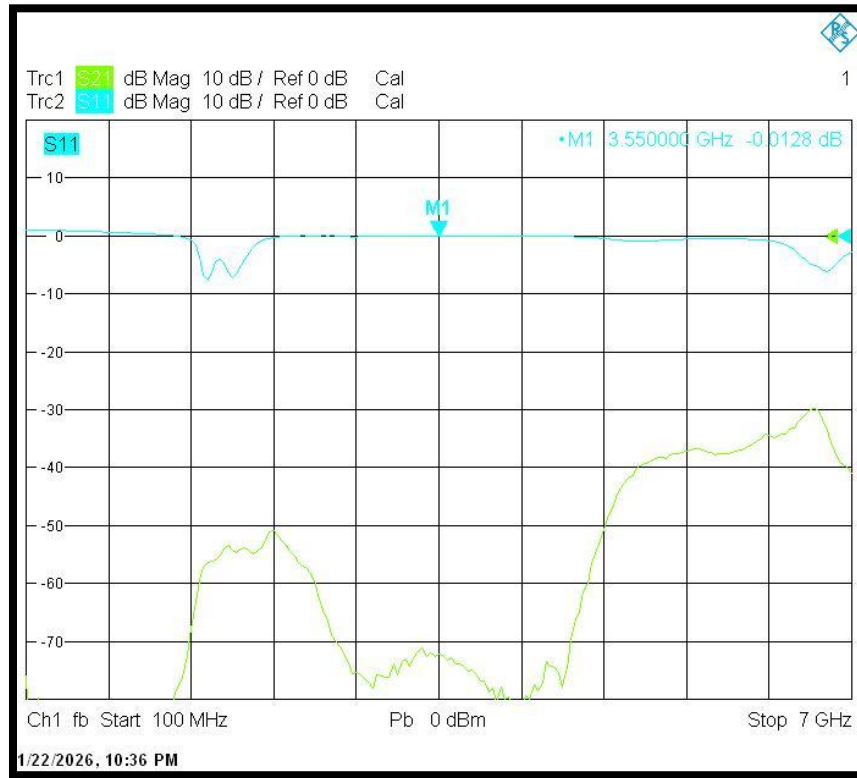


Figure 37: Result real testing

The figure 37 shows the measured of real testing result using a network analyzer, while the figure 25 is the simulation result from Genesys. In the measurement, the S21 response appears almost flat and close to 0 dB around 3.5 GHz, which means the signal is passing through the circuit with very low loss at the center frequency. The marker at 3.55 GHz shows about -0.01 dB, indicating good transmission at that point. However, the measured S11 response does not show clear, deep ripples like in the simulation, and instead looks smoother and less structured.

The primary cause of the observed non-similarity of the measured graph and the simulated one is that real hardware is never really perfect, whereas simulation is based on perfect conditions. In Genesys, the model is based upon the precise dimensions, perfect copper edges, ideal substrate attributes, and ideal port. Even in the actual fabricated circuit, there are tiny fabrication errors, and particularly in the narrow coupling gaps, the coupling strength between lines. This decreases the acuity of the resonances, which leads to the ripple pattern observed in simulation to be either weaker or smeared over in measurement.

Some other critical causes include the losses, which cannot be fully simulated, including conductor loss, surface roughness, dielectric loss, connector loss and soldering effects. These are losses that reduce the resonant behavior of the coupled lines, thus the deep S11 nulls and a series of ripples are significantly less noticeable in the measured result. Consequently, the response obtained appears more smooth and flatter than the Genesys graph.

Moreover, there are measurement setup effects that are also involved. The measured response can be disturbed by imperfect calibration, cable motion, connector misalignment and radiation by open microstrip structures. The effects are minor in nature but when they are added together they cause the ripple structure to become much less visible in the actuality than it does in simulation.

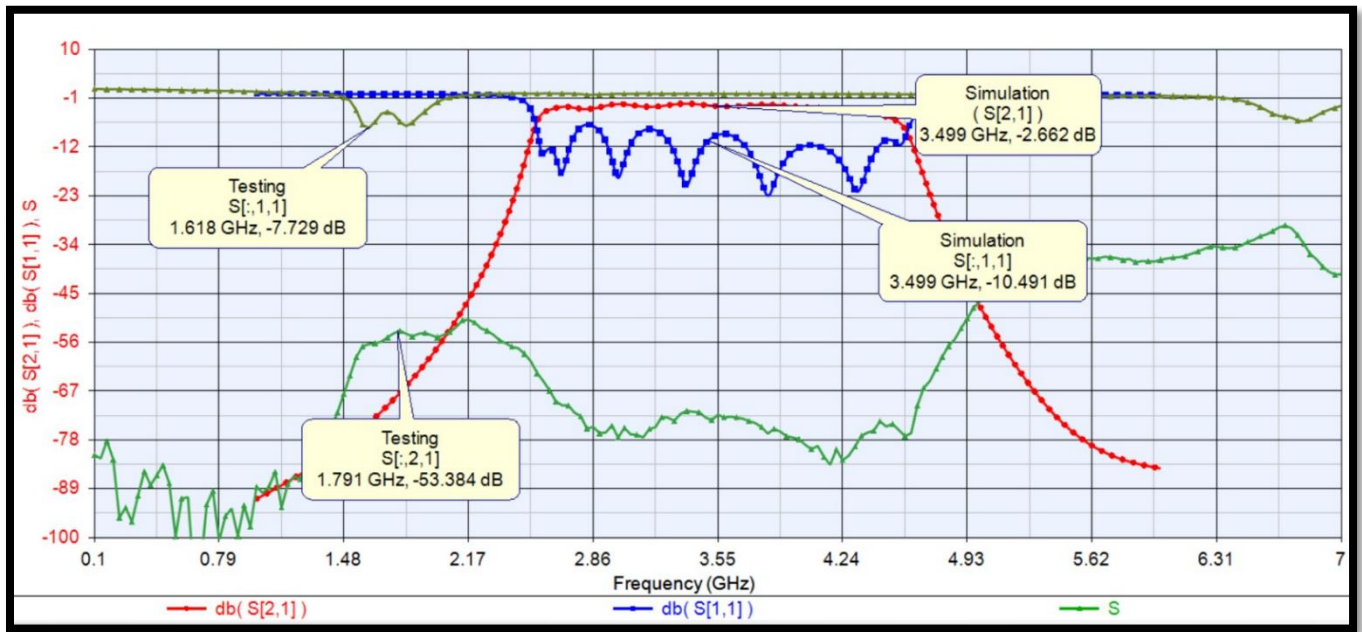


Figure 38: Result comparison between simulation and real testing in same graph

Figure 38 shows a direct comparison between simulation (Genesys) and real testing results plotted on the same graph, allowing the performance differences to be clearly observed. In the simulation, the S21 response (red curve) forms a well-defined passband between approximately 3.1 GHz and 3.9 GHz, with an insertion loss of about -2.66 dB at 3.499 GHz, indicating efficient signal transmission in the ideal modeled condition. At the same frequency, the simulated S11 response (blue curve) reaches around -10.49

dB, showing acceptable impedance matching and the intended ripple behavior produced by the coupled-line structure.

In contrast, the measured (testing) results, shown by the green curve, behave differently. The measured S21 at lower frequencies (for example, around 1.79 GHz) shows very high attenuation of about -53.38 dB, confirming good stopband rejection but also indicating that the measured response is more lossy and less sharply defined than in simulation. Similarly, the measured S11 at around 1.62 GHz is only about -7.73 dB, which is much higher than the deep nulls predicted by simulation. This indicates weaker impedance matching in the real circuit compared to the ideal simulated model.

These differences are largely explained by the fact that in simulation, it is assumed that the system is perfect with no fabrication losses, conductors, and dielectric loss, and the connectors are not lossy, whereas the real circuit is subject to fabrication loss, connector and soldering effects, and roughness of surfaces. Any little variation in the small gaps between the couplings has a strong effect on the coupling strength that destroys the ripple structure and alters the resonant frequencies. Because of this, the sharp ripples and deep nulls of the simulated S11 and deep nulls are less sharp and pronounced in the measured outcome.

Also, other sources of distortion to the measured response are measurement set-up issues such as calibration errors, cable loss, connector and connector mismatch, and radiation due to the untidied open microstrip lines. Such practical impacts both decrease the quality factor of the resonances and lead to frequency shifts and difference in amplitudes between simulation and measurement. Broadly speaking, Figure 38 shows that the general passband behaviour and the stopband rejection trend remain the same when comparing the ideal simulation environment with real hardware testing but that significant changes in ripple depth, matching level and precise frequency location are to be desired and to be expected in the actual hardware testing.

4.0 CONCLUSION

Abandoning any band, a 3.5 GHz coupled microstrip line bandpass filter was successfully designed, simulated, fabricated, and tested in this project. Design was initiated by simulation on Genesys in which the initial automatic optimization could not completely satisfy the design specifications, specifically in passband ripple and passive impedance match. This led to manual tuning, which emphasized on key physical parameters including line length (L), width (W) and coupling gaps (S1-S8). It was found that altering the length causes the passband to shift left or right and that alteration in the spacing of the gaps has a significant influence on the depth of the ripple and alteration in width leads to minimal changes in the frequency and magnitude. With this manual technique, a passband was obtained between 3.1 GHz and 3.9 GHz with seven ripples, which met the design requirement and performed higher than automatic optimization.

The artificial filter plan was then implemented on a microstrip substrate having total dimensions of 40mm x 11.11mm. Comparison of the simulation and physical measurements of dimensions revealed that a number of parameters particularly the small coupling gaps had a percentage error greater than +5%. This was not a result of design errors, but rather a result of PCB fabrication tolerances, copper etching effects and measurement limits which are far more significant in very small sizes. Larger parameters including the main width and length exhibited significantly smaller errors, and it was established that the quality of the fabrication could be used in microwave circuits with practical expectations of the quality.

In comparing the measured S-parameters to the simulated response, it was observed that the overall passband location and stopband behavior was similar, but the sharp ripple patterns and deep nulls as evident in the simulation were not so apparent in the measured response. The primary cause of this difference is due to real-world losses, such as dielectric loss, conductor loss, surface roughness, connector and soldering effects, and imperfect calibration of the measurement. These losses give the resonances less quality factor, flattening the response, and lowering the structure of ripple than the hypothetical Genesys model.

Generally, this paper has shown that simulation is a perfect reference whereas the measurement is an actual limitation. The ultimate constructed filter, in spite of these inevitable disparities, reached the desired working band and was about 3.5 GHz, satisfied the ripple criterion, and demonstrated reasonable correspondence with the simulation trends. The project has done quite well in bringing out the significance of manual tuning, fabrication tolerance

knowledge, and realistic expectation in changing simulation to hardware that is critical skills in actual microwave and RF filters design.

Project/Assignment Evaluation Form (50%)

CO1: Demonstrate the scattering parameters of a microwave circuit.

CO2: Convert lumped circuit elements into microstrip form.

CO3: Design microwave amplifier and microwave filter or oscillator, and verify one of the designs through fabrication.

Project/ Assignment:	Students' Names:
	1. MUHAMAD AKMAL HARITH BIN MOHD BAKTHIAR
	2. SITI AFIQAH BINTI MUSA
	3.
	4.

PART A

PO2 Identify, formulate, research literature and analyze complex electronics engineering problems reaching substantiated conclusions.								
Indicator (Domain)	Complex Engineering Problem (WP/EA)	Marks Allocated	Marks (M)					Weighted Marks Obtained
			1	2	3	4	5	
Identify and analyze detailed procedures for the calculations.	NA	5	Show minimal ability to identify and analyze detailed procedures for the calculations.	Show some ability to identify and analyze detailed procedures for the calculations.	Show adequate ability to identify and analyze detailed procedures for the calculations.	Show good ability to identify and analyze detailed procedures for the calculations.	Show excellent ability to identify and analyze detailed procedures for the calculations.	(=M X 1)
Formulate the parameters involved using appropriate calculations and according to the required design objectives.	NA	5	Show minimal ability to formulate the parameters involved using appropriate calculation and not according to design objectives	Show some ability to formulate the parameters involved using appropriate calculation and not according to design objectives	Show adequate ability to formulate the parameters involved using appropriate calculation and according to design objectives	Show good ability to formulate the parameters involved using appropriate calculation and according to design objectives	Show excellent ability to formulate the parameters involved using appropriate calculation and according to design objectives	(=M X 1)

Total (PART A)	/10
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PART B

PO3 Ability to design solutions for complex electronics engineering problems with appropriate consideration for public health and safety, culture, society, and environment.								
Indicator (Domain)	Complex Engineering Problem (WP/EA)	Marks Allocated	Marks (M)					Weighted Marks Obtained
			1	2	3	4	5	
Analyze the lumped-to-distributed element transformation scheme relevant to the design.	NA	10	Show minimal ability to analyze the lumped-to-distributed element transformation scheme.	Show some ability to -analyze the lumped-to-distributed element transformation scheme.	Show adequate ability to -analyze the lumped-to-distributed element transformation scheme.	Show good ability to -analyze the lumped-to-distributed element transformation scheme.	Show excellent ability to analyze the lumped-to-distributed element transformation scheme.	(=Mx2)
Analyze the transmission line theory and/or formulae involved relevant to the design.	NA	7.5	Show minimal ability to analyze the knowledge in transmission line theory and/or formulae involved.	Show some ability to analyze the knowledge and need to include transmission line theory and/or formulae involved.	Show adequate ability to analyze the knowledge and the need to include transmission line theory and/or formulas involved with minimal analysis.	Show good ability to analyze the knowledge and need to include transmission line theory and/or formulae involved with adequate analysis.	Show excellent ability to analyze the knowledge and need to include transmission line theory and/or formulae involved with comprehensive analysis.	(=Mx1.5)
Evaluate the design with reliable explanation that suits technological application	NA	2.5	Failed to evaluate the design with no explanation that suits the technological application.	Evaluate the design with an unreliable explanation that suits the technological application.	Evaluate the design with a reliable explanation that suits the technological application.	Evaluate the design with a good and reliable explanation that suits the technological application.	Evaluate the design with a comprehensive and reliable explanation that suits the technological application.	(=Mx0.5)

Total (PART B)	/20
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PART C

PO5 Ability to create, select, and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modeling, involving complex electronics engineering problems.								
Indicator (Domain)	Complex Engineering Problem (WP/EA)	Marks Allocated	Marks (M)					Weighted Marks Obtained
			1	2	3	4	5	
Explain the simulated design based on the calculations.	WP1, WP2, WP3	2.5	Show minimal ability to explain the simulated design based on the calculations.	Show some ability to explain the simulated design based on the calculations.	Show adequate ability to explain the simulated design based on the calculations.	Show a good ability to explain the simulated design based on the calculations.	Show an excellent ability to explain the simulated design based on the calculations.	(=Mx0.5)
Design the circuit layout according to the design objectives using modern engineering and IT tools with/without guidance	WP1, WP2, WP3	7.5	Show minimal ability to design the circuit layout according to the design objectives using modern engineering and IT tools with full guidance.	Show some ability to design the circuit layout according to the design objectives using modern engineering and IT tools with full guidance.	Show adequate ability to design the circuit layout according to the design objectives using modern engineering and IT tools with minimal guidance.	Show a good ability to design the circuit layout according to the design objectives using modern engineering and IT tools without guidance.	Show an excellent ability to design the circuit layout according to the design objectives using modern engineering and IT tools without guidance.	(=Mx1.5)
Specify and demonstrate the design with parametric evaluations.	WP1, WP2, WP3	7.5	Show minimal ability to specify and demonstrate the design with parametric evaluations.	Show some ability to specify and demonstrate the design with parametric evaluations.	Show adequate ability to specify and demonstrate the design with parametric evaluations.	Show a good ability to specify and demonstrate the design with parametric evaluations.	Show an excellent ability to specify and demonstrate the design with comprehensive parametric evaluations.	(=Mx1.5)

Adapt any deviation of the actual result from the target specification and include suggestions on how to mitigate it.	WP1, WP2, WP3	2.5	Show minimal adaptation towards the deviation in the results with required target specification and minimal suggestion to mitigate the design problem.	Show some adaptation towards the deviation in the results with required target specification and some suggestion to mitigate the design problem.	Show adequate adaptation towards the deviation in the results with required target specification and adequate suggestion to mitigate the design problem.	Show good adaptation towards the deviation in the results with required target specification and good suggestion to mitigate the design problem.	Show excellent adaptation towards the deviation in the results with required target specification and excellent suggestion to mitigate the design problem.	(=Mx0.5)
Total (PART C)								/20
TOTAL MARKS (PART A+PART B+PART C)								/50